# Redistricting with Endogenous Candidates*

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**Abstract**

I study partisan gerrymandering when district composition affects candidates' policy positions and voters' behavior. In the U.S., primary elections determine competing candidates, with districts' ideological composition shaping nominee selection. Thus redistricting affects not only which party wins but also candidates' ideologies. I find that classical gerrymandering strategies backfire when candidates emerge endogenously, particularly where extreme voters may select non-viable candidates. However, when properly designed to account for voter affiliation and preference intensity, gerrymandering becomes more powerful than traditional approaches. Using optimal transport theory, I characterize the optimal redistricting plan, which maximizes ideological distance between competing candidates. I analyze implications for Congressional polarization and minority representation.

Keywords: Gerrymandering, optimal transport, information design, polarization.

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# 1 Introduction

Partisan gerrymandering is the deliberate manipulation of electoral district boundaries to favor a political party. The 2012 Republican Redistricting Majority Project (REDMAP) is an example: despite receiving 1.4 million fewer votes than Democrats in U.S. House elections, Republicans secured a 33-seat majority through redistricting (Daley, 2020). Traditional approaches like REDMAP rely on presidential election data to classify areas as Republican or Democratic, implicitly assuming voter behavior remains unaffected by redistricting itself. The 2022 House elections offer an intriguing counterpoint: Republicans underperformed expectations, losing competitive districts they were favored to win, often where extreme candidates had emerged from primaries (Thakker, 2022; Rakich, 2022; Korte et al., 2022).

This paper studies optimal redistricting when candidates' policy positions and, consequently, voters' behavior, respond to the districts in place. I embed a two-stage model of elections at the district level into an optimal redistricting problem and illustrate how traditional gerrymandering can backfire when extreme voters select non-viable primary candidates. Using methods from optimal transport theory (Monge, 1781; Kantorovich, 1942), I show how savvy redistricters can create more powerful plans than standard redistricting allows for. Their strategy? Crafting district boundaries that drive a wedge between moderates and extremists in the opposing party, effectively turning their rivals' diversity into a liability[1].

**Illustrative examples.** To illustrate the forces in my model, it is useful to recall the standard redistricting setting. In the example in Figure 1, there's a U.S. state with two-thirds Democrats (in blue) and one-third Republicans (in red). A Republican redistricter has to partition voters into, say, three equipopulous districts, and wants to maximize the number of districts with a Republican majority. She applies the standard "pack-and-crack" technique. That is, she creates two "cracked" districts, District 1 and 2, consisting of just enough Republicans to have a majority, and one "packed" district,

[1] Redistricting can affect voter behavior through multiple channels. I focus on voters switching party when they dislike their nominee, but the key insight—that optimal gerrymandering exploits wedges between voter types—extends naturally to other mechanisms, like endogenous turnout. See more in Section 5.2.

made up of only Democrats. Thanks to gerrymandering, Republicans win two-thirds of districts with just one-third of overall Republican supporters. This is a classic result and one core point of the effects of gerrymandering.

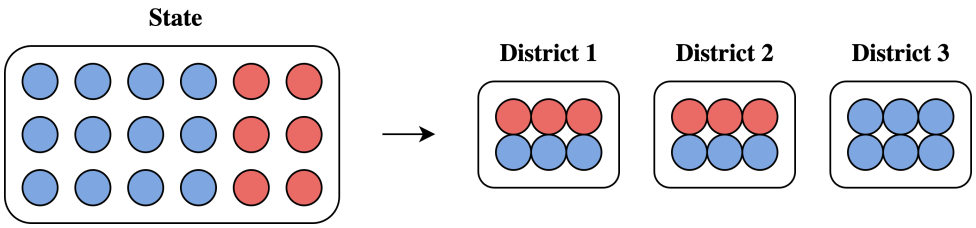



Figure 1: Standard pack-and-crack redistricting.

The approach above, however, assumes that party positions are fixed and exogenous. Suppose that party positions in each district are endogenous. Consider Figure 2. In the population, a small fraction of Democrats and an even smaller fraction of Republicans are moderates, represented by less intense shades of blue and red, respectively; all other voters are partisans (or extremists). Suppose primary elections precede the state-wide elections. In each district and for each party, primaries determine the candidate, who can be either a moderate or an extremist, depending on which voter type constitutes the majority within their party. In general elections, while partisans consistently vote for their party's candidate, moderate voters prefer a moderate candidate from *either* party over an extreme candidate from their own party. The standard "pack-and-crack" gerrymandering strategy can backfire. For example, consider District 1 and District 2 in Figure 2. In both districts, there is a majority of partisans among Republicans and a majority of moderates among Democrats. Hence, the Republican candidate caters to partisans, while the Democratic candidate caters to moderates. The only moderate Republican voter prefers voting for the moderate Democratic candidate rather than for a partisan Republican. Redistricting effectively converts the moderate Republican voter to a Democratic voter! Ultimately, with endogenous candidates, classical gerrymandering fails to achieve its goal: in this example, *all* districts end up selecting a Democratic candidate.

A real-life equivalent of such "dummymandering", as a bad gerrymandering is usually referred to in policy circles, can be found in the case of Oregon's fifth district: after Democrats redrew boundaries in anticipation of the 2022 mid-term elections, progressive Jamie McLeod-Skinner unexpectedly defeated seven-term centrist incumbent

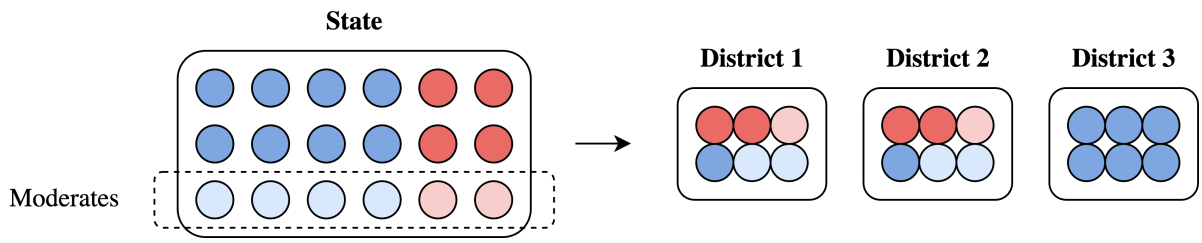



Figure 2: Pack-and-crack backfires.

Kurt Schrader in the Democratic primary. Crucially, McLeod-Skinner's victory hinged on a forty-point advantage in Deschutes County, which was only added to the district through a recent redistricting effort. This shift in Democratic candidate allowed Republican Lori Chavez-De Remer to appeal to moderate voters and flip the district for the first time since 1994 (Flaccus, 2022; Glueck, 2022; Scott and Weigel, 2022).[2]

Do such examples and real-life instances mean that gerrymandering is less powerful than previously thought? Quite the opposite. Consider a scenario where a Republican redistricter is aware of and exploits the endogeneity of voter behavior. Figure 3 illustrates an optimal gerrymandering plan. In District 1, despite having a Democratic majority, the composition is heterogeneous. There are sufficient Democratic partisans so that the Democratic candidate caters to their preferred policy, while the two moderate Democrats align more closely with the Republican candidate, who caters to the single moderate Republican voter in the district. Consequently, moderate Democrats prefer to vote Republican, even though they would have preferred a moderate Democratic candidate to a moderate Republican. As a result, Republicans win this district. District 3 consists of a majority of partisan Republicans, ensuring a straightforward Republican win. By strategically considering the endogeneity of candidates' selection and voters' behavior, the redistricter manages to secure *all districts* for the Republican party, despite Republicans constituting only one-third of the overall population. That is, gerrymandering can be more powerful than previously thought if its impact on voting behavior is taken into account.

[2] Another example is Ohio's ninth district, which was predicted to have R+3 to R+6 partisan lean, but ended up being won by the incumbent Democratic candidate by a thirteen points advantage (Lavelle, 2023).

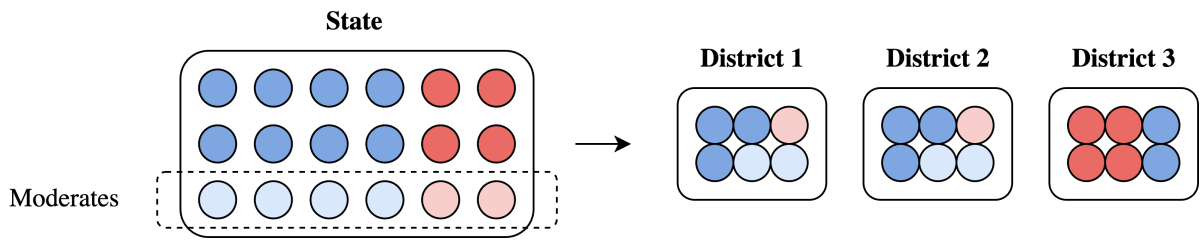



Figure 3: Gerrymandering is even more powerful than previously thought.

**Model.** There is a continuum of voters with single-peaked preferences over a unidimensional policy space. Voters are identified by their bliss point, which I sometimes refer to as their "type." Republicans are voters with bliss points above an exogenous cutoff, while Democrats are voters with bliss points below the cutoff.

A Republican designer partitions voters into equipopulous districts to maximize the number of districts won by Republicans.

The model incorporates a two-stage election process at the district level to determine voting behavior. First, in the primary elections, Democratic and Republican voters separately select their party candidates. Then, in the general election, all district voters choose between the two primary winners. Drawing on the work of Owen and Grofman (2006), I model electoral incentives so that primary candidates position themselves to appeal to their respective party medians.[3] To close the model, in districts lacking Republican or Democratic voters, I assume the corresponding party's candidate defaults to the most moderate position within their ideological spectrum. The general election outcome is decided by the district's median voter. The candidate whose position most closely aligns with the district median wins the election.

Given that redistricting in the U.S. takes place roughly every ten years, it is conceivable that redistricters cannot perfectly anticipate future populations' preferences at the time of redistricting. Therefore, I allow the designer to face uncertainty about voters' preferences, parametrized by a one-dimensional aggregate shock. In other words, after the designer commits to their redistricting plan, but before elections take place, a random shock occurs that uniformly shifts all voters' preferences by the same amount.

[3]In the Appendix, I provide microfoundations. The strict dependence on party medians is not necessary for my results: the driving force behind the model is responsiveness of candidates to party preferences. In Section 5.1, I provide an extension where candidates' positions are determined by a general quantile of the preference distribution within each party.

**Results.** In Section 2, I show that the problem can be simplified by restricting attention to a specific subset of feasible plans. Proposition 1 establishes that each optimal district must contain exactly two voter types: one type from below the population median and one type from above it, in equal proportions.

The intuition for this result emerges from the examples above. First, Figure 2 demonstrates how pack-and-crack backfires when District 1 mixes moderate and extreme Republican types. The designer can perfectly hedge against this possibility by creating cracked districts with no more than one Republican type, ensuring the optimal plan performs at least as well as standard pack-and-crack. Second, Figure 3 reveals a novel opportunity: In Districts 1 and 2, the ideological divide between moderate and extreme Democrats allows Republicans to win despite being in the minority. This same mechanism extends to districts packed exclusively with Democrats, where a moderate Republican candidate can peel off enough moderate Democratic voters to secure victory. To maximize this advantage, each such district should contain exactly two voter types: a higher type serving as moderate Democrat and a lower type as extreme Democrat. This hybrid strategy—packing Democrats while cracking them between moderates and extremists—allows the optimal plan to strictly outperform traditional pack-and-crack.

The effectiveness of gerrymandering depends critically on voter type differences within a district. The larger the wedge between the two voter types, the higher the probability of winning the district, making districts with significant gaps "safer" than more homogeneous ones. While the designer would ideally maximize this gap for all districts, she is constrained by the overall distribution of voter types. She therefore has to decide which voter types to allocate to safer, more heterogeneous, districts, and which to less safe, more homogeneous, districts.

As Theorem 1 summarizes, this task can be formalized as a Monge-Kantorovich optimal transport problem, where the designer determines how to optimally "transport" voter types from above the population median to match with types below it. The solution to this problem depends on the distribution of uncertainty about voter preferences.

In Section 3, I characterize such solution under the assumption that the aggre-

gate shock is S-shaped and symmetric around zero.[4] Proposition 3 demonstrates that there is a unique optimal redistricting plan. Voters far from the population median are matched in a "positive assortative" or "comonotonic" manner: high types above the median are paired with high types below it. However, for voters close to the median, the matching reverses to "negative assortative" or "anti-comonotonic": high types above the median are paired with low types below it.

The intuition is as follows. The districts most likely to be won are those with large ideological gaps between their two voter types, which requires matching voters from the right tail of the distribution with voters from the left tail. When working with these extreme voters, there's no benefit in creating an excessively large gap in one district at the expense of another. Instead, the designer uses positive assortative matching to maintain consistent, sufficient gaps across all such districts. However, closer to the median, ideological gaps become harder to create and districts become less likely to be won. Here, the designer switches to negative assortative matching: creating some very safe districts with large gaps at the expense of other districts that are given up as lost.

The symmetry assumption on the shock allows me to further determine which voters are matched positively versus negatively assortatively across the distribution. Proposition 4 serves as the analogue of a tangency condition in my framework: whenever the designer can reduce the mass of negatively assortative matches and redistribute it to positive assortative matches—where extreme types above the median pair with moderate types below it—he will do so.

This characterization yields clear comparative statics. For instance, when the distribution of voters is sufficiently "spread-out", the Republican redistricter benefits by more easily exploiting the wedge between moderate Democrats and extreme Democrats.

**Implications.** In Section 4, I explore implications of my gerrymandering model that are pertinent to current political and legal debates.

First, I examine how optimal gerrymandering strategies can exacerbate polarization in the U.S. House of Representatives. Proposition 6 predicts a notable ideological gap among elected representatives, who are either moderate Republicans or extreme

[4]The cumulative distribution function (CDF) of the shock is convex below zero and concave above it. Prominent cases of symmetric noise considered in the literature—e.g., normally-distributed noise—satisfy this assumption.

Democrats. Notably, extreme Democrats emerge in districts with a significant share of moderate voters, challenging the conventional view that political extremism stems from segregating extreme voters into homogeneous districts.

Second, I discuss the implications for minority representation in congressional districts. In "minority opportunity" districts, where minorities make up 40-50% of voters, minority candidates' success depends critically on white voter support. My model suggests this dependence can trigger "white backlash," where white voters unite against minority-preferred candidates, benefiting Republican candidates as predicted by my framework.

## 1.1 Related Literature

**Optimal partisan gerrymandering.** This paper primarily relates to the economic theory literature on optimal partisan gerrymandering, starting with Owen and Grofman (1988). While their work focuses on a binary voter type model, subsequent research has emphasized the importance of considering voters' preference intensities in redistricting strategies. In existing models, preference intensities play a role mostly due to uncertainty about voter preferences, typically assuming moderate voters are more susceptible to preference swings. Friedman and Holden (2008) show that with significant aggregate preference uncertainty, the optimal redistricting plan assigns extreme supporting voters to the same districts as extreme opposing voters, effectively neutralizing the latter's influence. The authors term this approach "matching slices," which is reminiscent of what I term negative assortative matching. Kolotilin and Wolitzky (2024) develop a more comprehensive model of gerrymandering that allows for both aggregate preference uncertainty and substantial shocks that affect voters independently, namely idiosyncratic uncertainty. Their findings suggest that when idiosyncratic uncertainty dominates, the optimal redistricting involves segregating the most extreme opposing voters while matching the remaining voters in a negative assortative manner. Competition between two parties, each controlling redistricting in distinct areas, is examined by Gul and Pesendorfer (2010). In their optimal plan, opposing voters are segregated, and more favorable voters are all pooled together.

I contribute to this literature by allowing individual voters' behavior to depend on the districts in place. One implication of this approach is that it results in a different

treatment of extreme opposing voters. While previous literature either predicts such extreme voters to be segregated or matched with extreme supporters, my model exploits them to turn moderate voters against their own party. To do so, the optimal plan prescribes "mis-matched slices," on either side of the preference distribution, resulting in novel implications.

**Information design.** As Kolotilin and Wolitzky (2024) show, the gerrymandering problem can be mapped onto an information design problem. The distribution of voter preferences serves as the "prior," districts function as "posteriors," and a redistricting plan represents a distribution of districts that satisfies a constraint, which is mathematically equivalent to Bayes plausibility. Indeed, in the special case of exogenous policies, my model becomes a variant of Bayesian persuasion (Kamenica and Gentzkow, 2011) for medians, which can be solved with recent off-the-shelf tools (Yang and Zentefis, 2024), as I show in Section 2. My solution sheds light on information design problems where payoffs depend on more intricate aspects of posterior distributions, such as the relative positions of conditional medians, rather than single summary statistics like means or medians.

**Optimal transport.** To characterize the solution, I leverage a Monge-Kantorovich optimal transport representation of the redistricter's problem (Monge, 1781; Kantorovich, 1942). Drawing on results from Chiappori et al. (2010) and Santambrogio (2015), I establish the existence and uniqueness of the solution and characterize it for key benchmark cases. Furthermore, I characterize the solution to an optimal transport problem where the surplus function is symmetric and S-shaped, thus quite different from the typical functional forms in the optimal transport literature. As far as I know, the only other papers that establish explicit properties of the solution for a concrete optimal transport problem with a non-typical surplus function are Boerma et al. (2023) and Strack and Yang (2025).

**Other topics in gerrymandering.** The broader literature on gerrymandering tackles a variety of different issues. The effects of redistricting on policy choice are considered by Shotts (2002) and Besley and Preston (2007), while the impact of gerrymandering on polarization in the House of Representatives is addressed, for instance, by McCarty

et al. (2009). Other important topics in redistricting that I do not explore in this paper relate to: including geographic constraints on gerrymandering (Puppe and Tasnádi, 2009), accounting for differential voter turnout (Bouton et al., 2023), and measures of electoral maldistricting (Gomberg et al., 2023).

# 2 The model

## 2.1 Setup and Statement of the Problem

**Voters and Parties.** Consider a continuum of voters with single-peaked preferences over uni-dimensional policy space $[\underline{v}, \overline{v}]$, a closed interval on the real line $\mathbb{R}$. A voter's ideal point, sometimes referred to as her type, is denoted by $v \in [\underline{v}, \overline{v}]$, with population distribution $\phi \in \Delta([\underline{v}, \overline{v}])$[5], assumed to have a strictly increasing and continuously differentiable cumulative distribution function (CDF), $F$, on the interior of $[\underline{v}, \overline{v}]$. I normalize the median of $F$ to be $v^m = 0$. Moreover, I assume voters' preferences to be symmetric around their ideal points[6].

There are two parties, the Democratic and the Republican party. Party affiliation is determined by a threshold $k \in [\underline{v}, \overline{v}]$. For simplicity, I assume $[\underline{v}, \overline{v}]$ is big enough, that is $\underline{v} < -k$. In particular, I call Republicans the voters such that $v \geqslant k$ and Democrats the voters such that $v < k$.

**Gerrymandering.** A Republican designer, or redistricter, is in charge of creating equipopulous districts so as to maximize his party's seat share. She allocates voters among a continuum of districts based on their type $v$, thus determining the distribution $\pi \in \Delta([\underline{v}, \overline{v}])$, with CDF $P$, of voter types within a district. A redistricting plan $\mathcal{H} \in \Delta(\Delta([\underline{v}, \overline{v}]))$[7] is, therefore, a distribution over distributions: it specifies the measure of districts with each distribution $\pi$ of voter types. To satisfy the equipopulous requirement typical of gerrymandering, any plan must be such that the following budget constraint holds[8]:

$$\int \pi d\mathcal{H}(\pi) = \phi. \tag{BC}$$

For instance, uniform redistricting imposes all districts to be the same, thus $\mathcal{H}(\phi) = 1$, while perfect segregation imposes that each voter type $v \in [\underline{v}, \overline{v}]$ constitutes a district on their own.

[5]For any complete and separable metric space $X$, I let $\Delta(X)$ denote the set of Borel probability measures on $X$, endowed with the weak* topology.

[6]For instance, a voter type $v$'s preference for policy $x \in [\underline{v}, \overline{v}]$ could be represented by utility function $u(x; v) = -|x - v|$.

[7]Note that $\Delta([\underline{v}, \overline{v}]])$ is metrizable as a complete and separable metric space with the weak* topology.

[8]The integral sign is used to denote the Lebesgue integral as defined in Aliprantis and Border (1994).

**Vote Shares.** In any district with distribution $\pi \in \text{supp}(\mathcal{H})$[9], two-stage elections are held. In primary elections, Republican voters choose a Republican candidate ($R$), while Democratic voters choose a Democratic candidate ($D$). During general elections, each voter chooses the candidate whose policy position is closer to their ideal point. The designer wins a district if and only if the Republican candidate receives at least a majority of the district vote.

I model electoral incentives so that $D$ and $R$ locate at the lowest median of $\pi(\cdot|v < k)$[10](the median of Democratic affiliates) and at the highest median of $\pi(\cdot|v \geqslant k)$ (the median of Republican affiliates)[11], respectively. There are various reasons why primary candidates might cater to their respective party medians rather than converge to the overall district median. In the Appendix, I provide a detailed model based on Owen and Grofman (2006), where both voters and candidates are uncertain about the position of the population median. However, it is important to note that the strict dependence on party medians is not necessary for my results to go through. For instance, the results are robust to any alternative rule that determines the position of candidate $D$ (respectively, $R$) through a convex combination of the median of Democratic (respectively, Republican) affiliates and the population median.

To close the model, I assume that whenever $\text{supp}(\pi) \subseteq [k, \infty)$ (respectively, $\text{supp}(\pi) \subseteq (-\infty, k]$), $D$ (respectively, $R$) takes position $k$. This assumption is equivalent to stating that even in a district with a very high Democratic (respectively, Republican) majority, there is always an arbitrarily small fraction of moderate Republicans (respectively, Democrats) close to $k$.

Formally, call $c_{\pi,D}$ and $c_{\pi,R}$ the position taken by $D$ and $R$ in district $\pi$. Then:

$$(c_{\pi,D}, c_{\pi,R}) = \begin{cases} (v^m_{\pi,D}, k) & if \ \text{supp}(\pi) \subseteq (-\infty, k] \\ (k, v^m_{\pi,R}) & if \ \text{supp}(\pi) \subseteq [k, \infty) \\ (v^m_{\pi,D}, v^m_{\pi,R}) & otherwise \end{cases},$$

[9]Throughout, $\text{supp}(\cdot)$ denotes the support of a probability measure, defined as the set of all points whose every open neighborhood has positive measure.

[10]Throughout, $\mu(Y|X)$ denotes the conditional probability distribution of $Y$ given $X$ according to measure $\mu$.

[11]The tie-breaking rule is chosen so as to ensure the existence of an optimum.

where:

$$v^m_{\pi,D} = \inf_a \quad \left\{a: \ \pi([a,\overline{v}]|v<k) \geqslant \frac{1}{2}\right\} \cap \left\{a: \ \pi([\underline{v},a]|v<k) \geqslant \frac{1}{2}\right\}$$
$$v^m_{\pi,R} = \sup_a \quad \left\{a: \ \pi([a,\overline{v}]|v\geqslant k) \geqslant \frac{1}{2}\right\} \cap \left\{a: \ \pi([\underline{v},a]|v\geqslant k) \geqslant \frac{1}{2}\right\}.$$

Voters choose their district representative by majority rule. Hence, the candidate who wins the elections is the one closer to the district's median. Formally, the position of the district representative, $c_\pi$, is:

$$c_\pi = \begin{cases} c_{\pi,D} & if \ \ v^m_\pi < \frac{c_{\pi,R}+c_{\pi,D}}{2} \\ c_{\pi,R} & if \ \ v^m_\pi \geqslant \frac{c_{\pi,R}+c_{\pi,D}}{2} \end{cases}$$

where:

$$v^m_\pi = \sup_a \quad \left\{a: \ \pi([a,\overline{v}]) \geqslant \frac{1}{2}\right\} \cap \left\{a: \ \pi([\underline{v},a]) \geqslant \frac{1}{2}\right\}.$$

The designer wins district $\pi$ if the winning candidate is the Republican candidate; that is, if $c_\pi \geqslant k$. Given redistricting plan $\mathcal{H}$, the designer's seat share is:

$$\int \mathbb{1}\left(c_\pi \geqslant k\right) d\mathcal{H}(\pi) = \int \mathbb{1}\left(v^m_\pi - \frac{c_{\pi,R}+c_{\pi,D}}{2} \geqslant 0\right) d\mathcal{H}(\pi).$$

**Aggregate Uncertainty.** Suppose that, after the designer commits to a plan, but before candidates choose their positions, an aggregate location shock affects all voters[12]. Formally, each voter experiences a common shock $\omega \in \mathbb{R}$, so that her ideal point becomes $v - \omega$. Assume that $\omega$ has distribution $\gamma \in \Delta(\mathbb{R})$, with CDF $G$, assumed to be Lipschitz continuous[13] and strictly increasing on $[\underline{v}-\overline{v}, 2\overline{v}-2\underline{v}]$. Any shock $\omega$ induces a new preference distribution in each district. I call $\pi^\omega$ such induced distribution, with CDF $P^\omega = P(v+\omega)$.

[12] A shock to preferences is particularly relevant in my setup, given that redistricting opportunities usually present themselves only every ten years. In any case, my solution encompasses the case in which the shock is arbitrarily small, so as to approximate the solution in the absence of a shock.

[13] For instance, if $G$ is everywhere continuously differentiable, it satisfies the assumptions. I merely require Lipschitz continuity to allow for $G$ to have a set of non-differentiability points of at most Lebesgue measure zero.

**Redistricter's Problem.** The redistricter, or designer, wishes to maximize the expected seat share won by Republicans. Her problem is:

$$\max_{\mathcal{H}\in\Delta(\Delta([\underline{v},\overline{v}]))} \int\int \mathbb{1}\left(v^m_{\pi^\omega} - \frac{c_{\pi^\omega,D} + c_{\pi^\omega,R}}{2} \geqslant 0\right) d\mathcal{H}(\pi)d\gamma(\omega) \tag{RP}$$

$$s.t. \int \pi d\mathcal{H}(\pi) = \phi.$$

A redistricting plan $\mathcal{H}$ is *feasible* if it satisfies constraint (BC). It is *optimal* if it is a solution to (RP).

It is instructive to compare (RP) to a redistricting problem with exogenous policies, whose solution can be found with off-the-shelf tools from recent literature. Suppose that candidates' positions are fixed: $c_{\pi^\omega,R} = \overline{v}$, $c_{\pi^\omega,D} = \underline{v}$. The redistricter's problem under exogenous policies is:

$$\max_{\mathcal{H}\in\Delta(\Delta([\underline{v},\overline{v}]))} \int\int \mathbb{1}\left(v^m_{\pi^\omega} \geqslant k^\star\right) d\mathcal{H}(\pi)d\gamma(\omega) \tag{RPEx}$$

$$s.t. \int P d\mathcal{H}(\pi) = F.$$

where $k^\star = \frac{\underline{v}+\overline{v}}{2}$. The object of interest to the designer is an element of $\Delta(\Delta([\underline{v},\overline{v}]))$, a distribution over distributions. However, in the case of exogenous policies, the objective function only depends on district medians. Hence, the problem can be mapped to a maximization problem over elements of $\Delta([\underline{v},\overline{v}])$, which are much simpler objects to work with. The set of feasible distributions of medians can be characterized using Theorem 2 in Yang and Zentefis (2024). The transformed problem is:

$$\max_{\chi\in\Delta([\underline{v},\overline{v}])} \int\int \mathbb{1}\left(v - \omega \geqslant k^\star\right) d\chi(v)d\gamma(\omega)$$

$$s.t. \quad \max\{2F(v)-1, 0\} \leqslant X(v) \leqslant \min\{2F(v), 1\}, \text{ for all } v \in [\underline{v},\overline{v}],$$

where $X$ is the CDF associated with $\chi \in \Delta([\underline{v},\overline{v}])$. Switching the order of integration, it can be rewritten as:

$$\max_{\chi\in\Delta([\underline{v},\overline{v}])} \int G(v - k^\star) d\chi(v)$$

$$s.t. \quad \max\{2F(v)-1, 0\} \leqslant X(v) \leqslant \min\{2F(v), 1\}, \text{ for all } v \in [\underline{v},\overline{v}].$$

Since the problem is linear in $\chi$, it suffices to focus attention on the extreme points of the set of feasible distributions of medians. Again, Yang and Zentefis (2024) characterize such a set in Theorem 1. In my case, since $G$ is increasing, the solution is $X^\star = \max\{2F - 1, 0\}$, which dominates all other feasible distributions of medians, in the first-order-stochastic-dominance sense. We can now recover all solutions to (RPEx) as all those redistricting plans that induce $X^\star$. It is easy to see that there are many such plans. The following proposition, already proven with a different argument by Kolotilin and Wolitzky (2024), describes such plans and summarizes the result for exogenous policies.

**Proposition 0.** *A feasible redistricting plan $\mathcal{H} \in \Delta(\Delta([\underline{v}, \overline{v}]))$ is a solution to the redistricting problem with exogenous policies (RPEx) if and only if, for all $\pi \in \operatorname{supp}(\mathcal{H})$, except for at most a zero-measure subset, there exists $v_\pi \geqslant 0$ such that $\pi(\{v_\pi\}) = \pi(\{v : \ v \leqslant 0\}) = \frac{1}{2}$.*

In the case of endogenous policies, it is not possible to reduce the object of interest to a distribution over a uni-dimensional space, because the objective function depends both on district medians and on candidates' positions. Nevertheless, in the next subsection, I show that the problem can still be simplified and solved using a different set of tools. In particular, I show that the object of interest can be reduced to a joint distribution over two uni-dimensional spaces, thus invoking the literature on the so-called optimal transport.

## 2.2 The Designer's Problem as an Optimal Transport Problem

While the redistricter's problem (RP) appears challenging, it can be simplified and restated as an optimal transport problem. The following result provides necessary conditions for optimality and constitutes the main ingredient for this transformation.

**Proposition 1.** *A feasible redistricting plan $\mathcal{H} \in \Delta(\Delta([\underline{v}, \overline{v}]))$ is optimal only if, for all $\pi \in \operatorname{supp}(\mathcal{H})$, except for at most a zero-measure subset, there exist $v' \geqslant 0$ and $v'' \leqslant 0$ such that $\pi(\{v'\}) = \pi(\{v''\}) = \frac{1}{2}$.*

Proposition 1 constrains the set of feasible plans: optimality requires that the population of any district has at most two types. Moreover, each district in an optimal

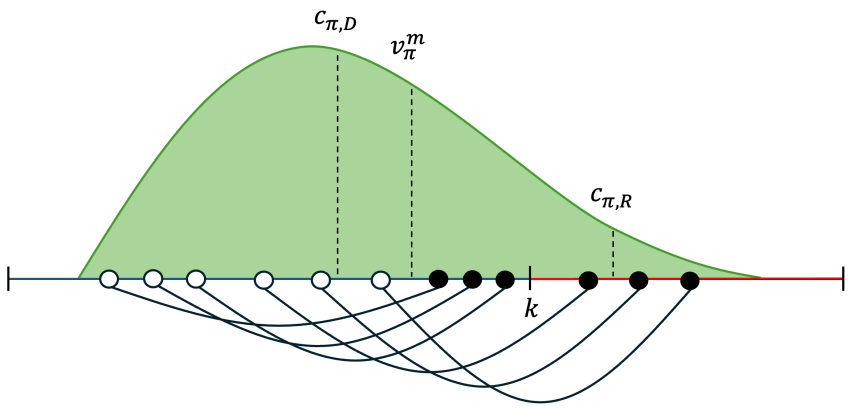
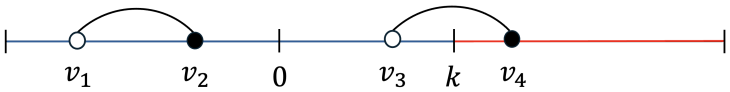


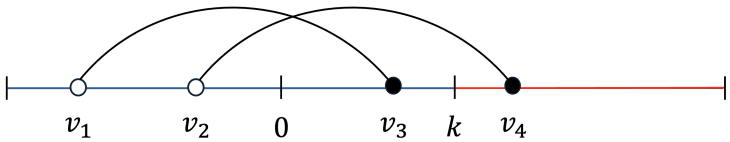

Figure 4: district $\pi$ (in green) can be replaced by a continuum of binary districts. Each such district matches types above the median of $\pi$ (●) to types below it (○) in a positive assortative manner.

plan must place half the mass on a voter type above the median of $F$ and the rest on a voter type below the median of $F$. Effectively, this proposition states that (RP) is a matching problem of voter types across the median of their population distribution. The proof proceeds in two steps, for which I now provide an intuition.

First, it can be shown that any district in an optimal plan must be a *binary district*. That is, it must contain at most two voter types (in equal proportions). Consider a plan $\mathcal{H}$ and any district $\pi$ in its support. As it turns out, $\mathcal{H}$ can be improved in the following way. Imagine replacing $\pi$ with multiple smaller districts (possibly infinitely many), where each new district contains at most two voter types in equal proportions: a high type from above the median $v_\pi^m$ and a low type from below $v_\pi^m$. Further, imagine that types above $v_\pi^m$ and types below $v_\pi^m$ are allocated to districts in a positive assortative manner. Specifically, suppose that any two districts contain types $(h, l)$ and $(h', l')$, respectively. If the first district has a higher high type, $h > h'$, then it must also have a higher low type, $l \geqslant l'$. Such construction is illustrated in Figure 4. For illustration purposes, suppose that the shock realization is $\omega = 0$. Then, district $\pi$ is lost, since $v_\pi^m$ is closer to the Democratic candidate, $c_{\pi,D}$, than it is to the Republican candidate, $c_{\pi,R}$. Nevertheless, each of the binary districts depicted in Figure 4 is won. Indeed, consider any of the districts $\hat{\pi}$ where the highest type (●) is above $k$. Then, the Republican candidate $c_{\hat{\pi},R}$ sits exactly at the median $v_{\hat{\pi}}^m$, so that $c_{\hat{\pi},R} = v_{\hat{\pi}}^m = \bullet$, and the district is won. Now consider any of the districts $\tilde{\pi}$ where the highest type is below $k$, so that $\tilde{\pi}$




(a) Two districts containing types from the same side of the population median.




(b) Alternative matching.

Figure 5: Districts must match voter types across the population median.

contains only Democrats. Then, the Republican candidate $c_{\tilde{\pi},R}$ sits at $k$, the Democratic candidate $c_{\tilde{\pi},D}$ sits at the lowest type (○), and the highest type (●) constitutes the median voter $v^m_{\tilde{\pi}}$. Note that $c_{\tilde{\pi},R}$ is more moderate than $c_{\pi,R}$, $c_{\tilde{\pi},D}$ is more extreme than $c_{\pi,D}$, and $v^m_{\tilde{\pi}}$ is higher than $v^m_{\pi}$, so that each $\tilde{\pi}$ is won. The fact that high types (●) are matched to low types (○) in a positive assortative manner ensures that the above reasoning works no matter the realization of the shock $\omega$.

Second, binary districts must contain exactly two voter types: the highest type drawn from above 0, the population median, and the lowest type drawn from below 0. Figure 5 illustrates this point. Suppose there are two districts, one containing types $v_4$ and $v_3$ and the other containing types $v_2$ and $v_1$, as shown in Figure 5a. Consider the alternative matching shown in Figure 5b, where $v_1$ is paired with $v_3$, while $v_2$ is paired with $v_4$. It can be shown that the pairing in 5b is superior to the one in 5a. For illustration purposes, suppose the realization of the shock is $\omega = 0$. Consider the district containing $v_4$. In both configurations, the Republican candidate sits exactly at $v_4$ and receives exactly half the votes, enough to win the district. Now consider the district containing $v_1$. In both cases, the Democratic candidate sits at $v_1$ and the Republican candidate sits at $k$. In 5a, $v_2$ is closer to $v_1$ than to $k$, so that all Democrats vote for the Democratic candidate and the district is lost. However, in 5b, type $v_3$ prefers voting for the Republican candidate, so that the district is won with exactly half the votes.

Proposition 1 justifies the definition of a relation between the set of redistricting plans and the set of joint distributions over $[\underline{v}, 0] \times [0, \overline{v}]$. Define $\phi' = \phi(\cdot|v \geqslant 0)$ and

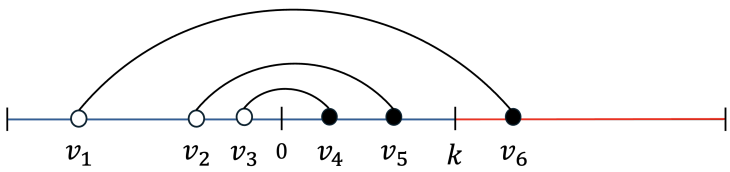



Figure 6: Of the three depicted districts, only the one containing voter types $v_3$ and $v_4$ is lost when $\omega = 0$.

$\phi'' = \phi(\cdot|v < 0)$. In words, $\phi'$ is the distribution of voter types conditional on them being above the population median, while $\phi''$ is the distribution of voter types conditional on them being below the population median. I call $T(\phi', \phi'') \subseteq \Delta([\underline{v}, 0] \times [0, \overline{v}])$ the set of joint distributions having marginals $\phi'$ and $\phi''$.[14] Following the literature on optimal transport, I sometimes refer to $T(\phi', \phi'')$ as the set of *transport plans* from $\phi'$ to $\phi''$.

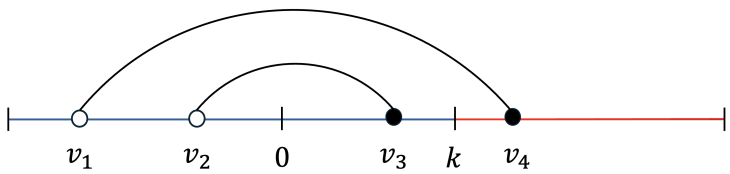
Call $\Delta_2 \subseteq \Delta(\Delta([\underline{v}, \overline{v}]))$ the set of feasible plans $\mathcal{H}$ such that, for all $\pi \in \text{supp}(\mathcal{H})$, there exist $v' \geqslant 0$ and $v'' \leqslant 0$, with $\pi(\{v'\}) = \pi(\{v''\}) = \frac{1}{2}$. As it turns out, there is a one-to-one map between $\Delta_2$ and $T(\phi', \phi'')$. Moreover, note that any district placing positive mass on $v' \geqslant 0$ and $v'' \leqslant 0$ can be won in one of two ways. For instance, after the realization of the location shock $\omega$, it can be that $v' - \omega$ is above $k$, so that the district has at least a majority of Republican affiliates. For such a district, I have $v^m_{\pi^\omega} = c_{\pi^\omega, R} = v' - \omega$ and $c_{\pi^\omega, D} = v'' - \omega$. Alternatively, it can be that $v' - \omega$ is less than $k$, but still closer to $k$ than it is to $v'' - \omega$, so that the district is split fifty-fifty between "extreme" and "moderate" Democrats, the latter choosing the default moderate Republican candidate. For such a district, I have $v^m_{\pi^\omega} = v' - \omega$, $c_{\pi^\omega, R} = k$, and $c_{\pi^\omega, D} = v'' - \omega$, with $k - (v' - \omega) \leqslant (v' - \omega) - (v'' - \omega)$. In Figure 6, three types of districts are shown. When $\omega = 0$, only the district containing voter types $v_3$ and $v_4$ is lost.

Given the above discussion, I can rewrite (RP) as:

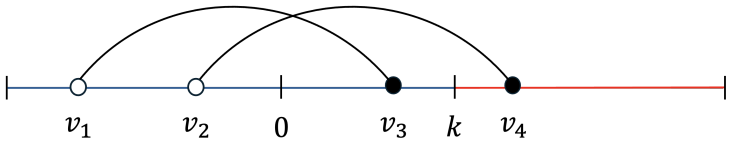
$$\max_{\tau \in T(\phi', \phi'')} \int\int \mathbb{1}(v' - \omega \geqslant k) + \mathbb{1}(v' - \omega < k)\mathbb{1}\left(v' - \omega - \frac{v'' - \omega}{2} \geqslant \frac{k}{2}\right) d\tau(v', v'') d\gamma(\omega).$$

By switching the order of integration and further manipulating the above, I get the

[14] Formally, $T(\phi', \phi'') = \left\{\tau \in \Delta([\underline{v}, 0] \times [0, \overline{v}]) : \text{proj}_{[\underline{v}, 0]} \# \tau = \phi'', \text{proj}_{[0, \overline{v}]} \# \tau = \phi'\right\}$, where: $\text{proj}_{[\underline{v}, 0]}$ and $\text{proj}_{[0, \overline{v}]}$ denote the projection functions of $[\underline{v}, 0] \times [0, \overline{v}]$ on $[\underline{v}, 0]$ and $[0, \overline{v}]$, respectively; $(\text{proj}_{[\underline{v}, 0]} \# \tau)(A) = \tau(\text{proj}^{-1}_{[\underline{v}, 0]}(A))$ and $(\text{proj}_{[0, \overline{v}]} \# \tau)(A) = \tau(\text{proj}^{-1}_{[0, \overline{v}]}(A))$ for all $A$ measurable.




(a) The district containing voter types $v_1$, $v_4$ is considerably safer than the one containing types $v_2$, $v_3$.




(b) The two districts are of similar safety.

Figure 7: Two alternative configurations of districts.

optimal transport problem:

$$\max_{\tau \in T(\phi', \phi'')} \int G\left(2v' - v'' - k\right) d\tau(v', v''). \tag{OTP}$$

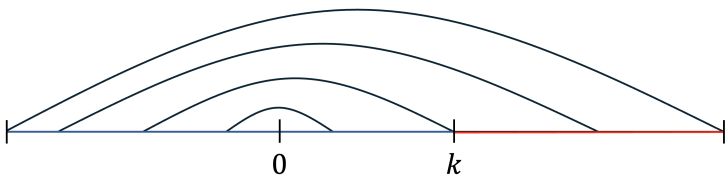
A transport plan $\tau$ in $T(\phi', \phi'')$ is *pure* whenever $(v', v'') \in \text{supp}(\tau)$ implies $(v', \tilde{v}''), (\tilde{v}', v'') \notin \text{supp}(\tau)$ for $v' \neq \tilde{v}'$ and $v'' \neq \tilde{v}''$. Intuitively, purity requires that no "splitting of masses" occurs across voter types. A pure plan is sometimes referred to as a *transport map*.

Define $T^\star \subseteq T(\phi', \phi'')$ as the set of solutions to (OTP) and $\Delta_2^\star \subseteq \Delta_2$ as the set of solutions to (RP) in $\Delta_2$. The optimal transport problem (OTP) is *equivalent* to the redistricter's problem (RP) if there exists a bijection from $T^\star$ to $\Delta_2^\star$. The following theorem summarizes the discussion in this subsection.

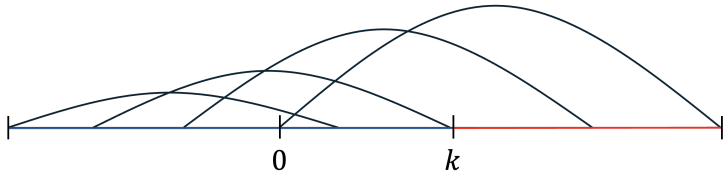
**Theorem 1.** *The optimal transport problem (OTP) is equivalent to the redistricter's problem (RP).*

# 3 Characterizing the Optimal Redistricting Plan

In this section, I characterize the solution(s) to the optimal transport problem (OTP) under different assumptions on the shock distribution $G$. As it turns out, the optimal redistricting plan depends heavily on the shape of $G$ and, in most cases, on the distribution of voters $F$.




(a) When $G$ is convex, the optimal transport plan maps types above 0 to types below 0 in a negative assortative manner.




(b) When $G$ is concave, the optimal transport plan maps types above 0 to types below 0 in a positive assortative manner.

Figure 8: The unique optimal plan under $G$ convex/concave.

## 3.1 Benchmark Cases

I start by describing a few benchmark cases before focusing on the most realistic case of a symmetric, S-shaped shock. Suppose that $G$ is strictly convex on its support. The location shock "shifts" the distribution of preferences. A negative shock shifts it to the right, increasing the fraction of Republican voters (hence it is a favorable shock), while a positive shock shifts the distribution to the left (so it is unfavorable). When $G$ is convex, the marginal benefit of slightly improving a competitive district's safety is outweighed by the cost of marginally reducing the safety of an already secure district. Consider Figure 7. Panel 7a depicts two districts, the one containing voter types $v_1$, $v_4$ considerably safer than the one containing types $v_2$, $v_3$. One can think of constructing two alternative districts of intermediate safety, as in panel 7b, by matching $v_1$ to $v_3$ and $v_2$ to $v_4$. Because $G$ is convex, the designer always prefers the matching in *7a* to the one in *7b*. In other words, the designer wants to create very safe districts consisting of extremists of both parties, alongside very unsafe districts consisting of more moderate voters. Based on this intuition, it can be shown that there exists a unique, pure solution to OTP and that such solution maps types above zero to types below zero in a *negative assortative manner*[15]. Figure 8a illustrates such negative assortative map.

[15]Formally, a *negative assortative map* $\tau \in T(\phi', \phi'')$ is such that, for all $(v', v''), (\tilde{v}', \tilde{v}'') \in \text{supp}(\tau)$, $v' > \tilde{v}'$ implies $v'' \leqslant \tilde{v}''$.

**Concave Shock.** The case of a strictly concave $G$ is analogous to that of a strictly convex $G$. In this case, the marginal benefit of slightly improving a competitive district's safety exceeds the cost of marginally reducing the safety of an already secure district. Hence, the designer prefers the configuration in Figure 7b to the one in Figure 7a and tries to create districts of similar safety. This preference for more balanced districts creates a natural tendency against negative assortative matching. Indeed, it can be shown that there exists a unique, pure solution to OTP that maps types above 0 to types below 0 in a *positive assortative manner*[16], as illustrated in Figure 8b.

**Uniform Shock.** For the sake of completeness, I analyze the case when $G$ is affine on its support. It is easy to see that, given the linearity of $G$, problem OTP does not depend on $T$, therefore $T^\star = T(\phi', \phi'')$. In other words, any redistricting plan in $\Delta_2$ is optimal.

The following result formalizes the findings for benchmark cases.

**Proposition 2.** *Consider the following cases:*

- *If $G$ is strictly convex on its support, there exists a unique, pure solution to OTP and it is a negative assortative map.*

- *If $G$ is strictly concave on its support, there exists a unique, pure solution to OTP and it is a positive assortative map.*

- *If $G$ is affine on its support, any $\tau \in T(\phi', \phi'')$ is a solution to (OTP).*

To prove Proposition 2, I use results in the optimal transport literature as off-the-shelf tools. Such tools fail to be useful in cases where $G$ is neither convex nor concave. In the next subsections, I show how the above benchmark results can still be used as building blocks to characterize the solutions to (*OTP*) when the shock is symmetric, strictly convex below zero, and strictly concave above zero, or S-shaped.

[16] Formally, a *positive assortative map* $\tau \in T(\phi', \phi'')$ is such that, for all $(v', v''), (\tilde{v}', \tilde{v}'') \in \text{supp}(\tau), v' > \tilde{v}'$ implies $v'' \geqslant \tilde{v}''$.

## 3.2 S-shaped and Symmetric Shock

Suppose the shock is strictly S-shaped and symmetric around zero.[17] Any normal shock with mean zero falls under this category. For instance, a normal shock with sufficiently small variance is of particular interest because it approximates the solution to (RP) in the absence of an aggregate shock. As it turns out, (OTP) admits a unique solution. Building on the benchmark cases studied in the previous subsection, I show that such a solution is the convex combination of a positive assortative map and a negative assortative map, each over an appropriate subset of $[0,\overline{v}] \times [\underline{v},0]$. Figure 9a depicts $[0,\overline{v}] \times [\underline{v},0]$ and partitions it into two subsets. For any $(v',v'')$ belonging to the purple region, $2v' - v'' - k$ is less than zero, while for any $(v',v'')$ belonging to the green region, $2v' - v'' - k$ is greater than zero. Hence, $G$ is convex on the purple subset of $[0,\overline{v}] \times [\underline{v},0]$ and concave on the green one. Building on the logic established in the benchmark cases, I show that any two couples of voter types falling in the green subset of $[0,\overline{v}] \times [\underline{v},0]$ must constitute a positive assortment, in order to be part of a solution to (OTP). Similarly, any two couples of voter types falling in the purple subset of $[0,\overline{v}] \times [\underline{v},0]$ must constitute a negative assortment. The following result formalizes such intuition.

**Proposition 3.** *Suppose the aggregate shock is S-shaped and symmetric around* 0*. There exist* $\tau^+$, $\tau^- \in \Delta([0,\overline{v}] \times [\underline{v},0])$ *and* $\alpha \in [0,1]$ *such that:*

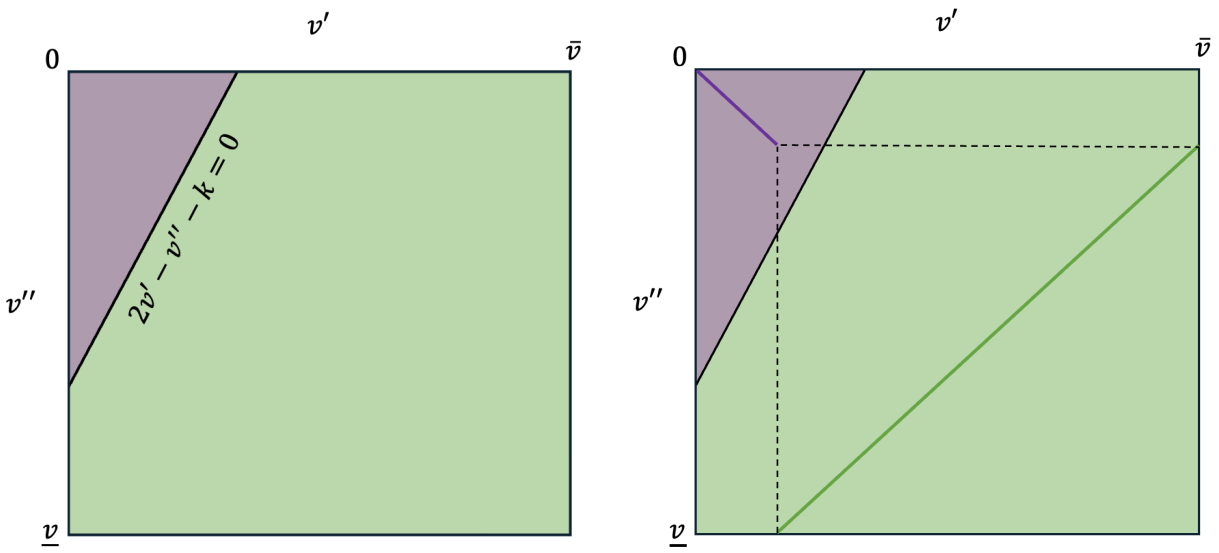
1. $\tau^-$ *is a pure, negative assortative map with path-connected* $\text{supp}(\tau^-) \subseteq \{(v',v'') \in [0,\overline{v}] \times [\underline{v},0] : 2v' - v'' - k \geqslant 0\}$

2. $\tau^+$ *is a pure, positive assortative map with path-connected* $\text{supp}(\tau^+) \subseteq \{(v',v'') \in [0,\overline{v}] \times [\underline{v},0] : 2v' - v'' - k \leqslant 0\}$

3. $(1-\alpha)\tau^- + \alpha\tau^+$ *is the unique solution to OTP.*

As an example, Figure 9b depicts the support of a pure candidate for optimality, when $\phi$ is uniform. In this case, the linearity of $F$ results in a linear mapping of voter types above zero and below zero, with negative slope for matches in the purple region and positive slope for matches in the green region.

[17] Formally, $G$ is strictly convex on $[\underline{v},0]$ and strictly concave on $[0,\overline{v}]$. Moreover, for all $x$, $G(x) = 1 - G(-x)$.

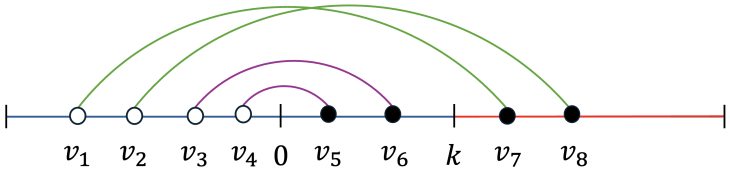



(a) The purple subset of $[\underline{v}, 0] \times [0, \overline{v}]$ hosts negative assortments, while the green subset hosts positive assortments.

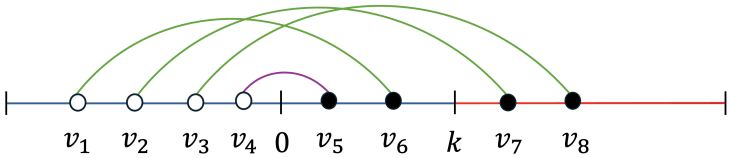
(b) Example of $\tau = \alpha\tau^- + (1-\alpha)\tau^+$ under $\phi$ uniform.

Figure 9: The unique solution to (OTP) is the convex combination of a negative assortative map and a positive assortative map.

Let $T^{\pm}$ be the set of all plans $\tau \in T(\phi', \phi'')$ consistent with Proposition 3. Having established that the unique solution to (OTP) must be an element of $T^{\pm}$, can it be further characterized? Consider Figure 10. Two plans, both belonging to $T^{\pm}$, are depicted. In Figure 10a, voter types $v_1$, $v_2$, $v_7$, and $v_8$ are matched positively assortatively, while $v_3$, $v_4$, $v_5$, and $v_6$ are matched negatively assortatively; in Figure 10b, voter types $v_1$, $v_2$, $v_3$, $v_6$, $v_7$ and $v_8$ are matched positively assortatively, while only $v_4$ and $v_5$ are matched negatively assortatively. The latter configuration is obtained from the former by breaking the match between $v_3$ and $v_6$ and including them in the positive assortative match. Which plan is preferred by the designer? On one hand, Figure 10b introduces an additional match where $2v' - v'' - k > 0$. On the other hand, it reduces the magnitude by which this inequality is satisfied by the other matches in green. As it turns out, the configuration in Figure 10b proves superior to the one in Figure 10a. This follows from the symmetry assumption of the shock, since $G$ is steeper around zero than at any positive value. This implies that the marginal benefit of converting a negative assortative match into a positive assortative one is larger than the marginal cost of reducing the degree of positivity in existing green matches (as long as they stay positive). This means that, for any $\tau$ in $T^{\pm}$, if I extract mass from $\tau^-$ and reallocate it to $\tau^+$ while




(a) Two positive assortative matched (in green) and two negative ones (in purple).




(b) Three positive assortative matched (in green) and one negative (in purple).

Figure 10: Under a symmetric shock, the configuration in the bottom panel is preferred to the one in the top panel.

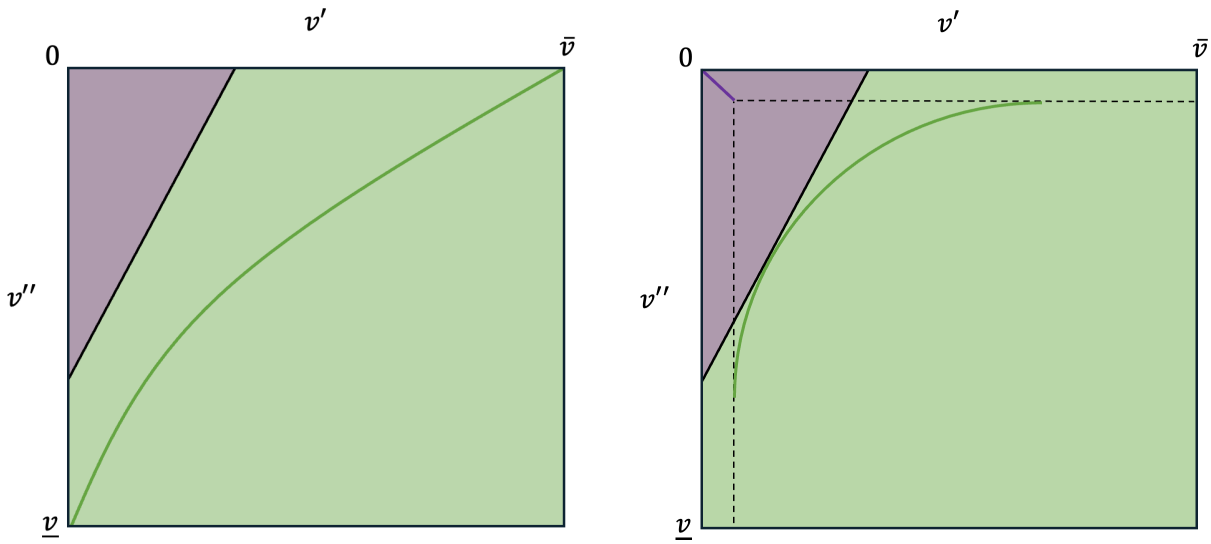
remaining $T^{\pm}$, the designer is better off (Lemma 7 in the Appendix formalizes this).

In turn, this implies that, at the optimum, either $\tau$ is fully positive assortative ($\alpha = 1$), or the support of its positive assortative part $\tau^+$ is tangent to the line $2v' - v'' - k = 0$. Otherwise, some mass could be moved to the positive assortative region, resulting in an improvement. Formally, for an element $\tau = \alpha\tau^+ + (1-\alpha)\tau^-$ belonging to $T^{\pm}$, define the most favorable match in $\tau^-$ to be $(v'_{max}, v''_{max}) = \arg\max_{\text{supp}(\tau^-)}(2v' - v'' - k)$. The following holds.

**Proposition 4.** *Suppose the shock is symmetric and S-shaped around* 0. *If* $\tau = \alpha\tau^+ + (1-\alpha)\tau^- \in T^{\pm}$ *is the solution to (OTP), there exists* $(v', v'') \in \text{supp}(\tau^+)$, *with* $v'' \leqslant v''_{max}$ *and* $v' \geqslant v'_{max}$, *such that:*

$$(1-\alpha)(2v' - v'' - k) = 0.$$

Figure 11 illustrates Proposition 4. When the solution to (OTP) is pure, the above result reduces to finding a unique "switching point", or cutoff, between the green and purple regions that minimizes the mass assigned to the purple region. However, the case of a plan that is not pure, meaning that some voter types might belong to both regions simultaneously, is likely. Figure 12 shows the simulated solution when both $F$ and $G$ are normal distributions, for different values of $k$. Areas of non-purity are




(a) The solution is fully positive assortative.

(b) Tangency condition.

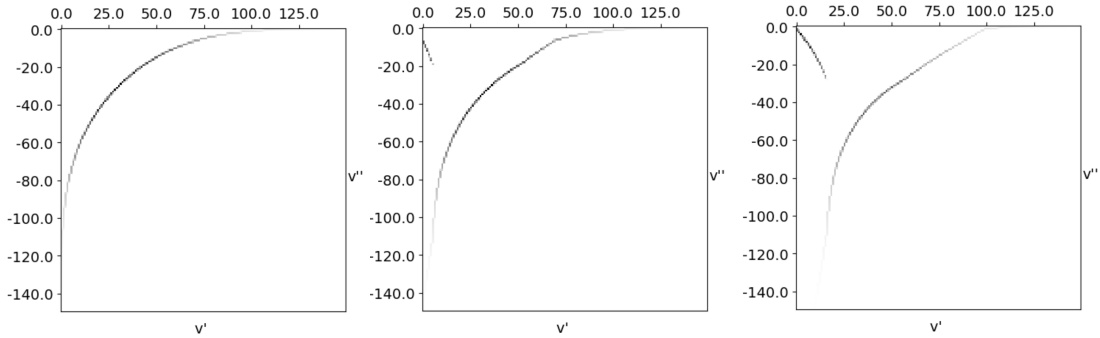
Figure 11: Under a symmetric shock, the solution is either fully positive assortative, or the positive assortative map touches $2v' - v'' - k = 0$.

evident. Note that as $k$ increases, and the fraction of Republicans decreases, more and more voters are matched in a negative assortative manner. This aligns with what Proposition 4 suggests. Specifically, when $k$ is low, the redistricter can create many districts satisfying $2v' - v'' - k > 0$ through positive assortative matching. However, as $k$ increases, the tangency condition binds, and this becomes harder to achieve, forcing the redistricter to resort to negative assortative matching to maintain as many winning districts as possible.

**Comparative Statics.** Cleear comparative statics can be obtained by manipulating the "spread" of the original distribution of voters. What happens if the tails of the distribution become "fatter" fatter, so that more extreme voters appear on both sides of the spectrum? More precisely, consider a new distribution of voters $\tilde{\phi}$, a median-preserving spread (Definition 5 in the Appendix) of the original $\phi$, and subject to the same assumptions. The following holds.

**Proposition 5.** *Suppose $\tilde{\phi}$ is a median-preserving spread of $\phi$. The expected seat-share won by the designer under $\tilde{\phi}$ is strictly larger than under $\phi$.*

In other words, the designer benefits from spreading out the ideological distribu-




(a) $k = 60$ (b) $k = 75$ (c) $k = 100$

Figure 12: Solution for simulated $F$ normal, $G$ normal, for $k = 60,\ 75,\ 100$

tion of voters by more easily exploiting the wedge between moderate Democrats and extreme Democrats.

# 4 Political and Legal Implications

Because of its effects on democratic representation and political polarization, gerrymandering has been controversial for decades. In this section, I discuss two implications of my model for current political and legal debates.

## 4.1 Gerrymandering and Congress Polarization

The relationship between gerrymandering and partisan polarization in the U.S. House of Representatives is more complex than public discourse suggests. Gerrymandering is often blamed for heightened partisanship and legislative gridlock, but empirical research remains divided. McCarty et al. (2009) question its significance, while others find measurable effects (Kenny et al., 2023). Theoretical models with exogenous candidates yield conflicting predictions. Models with segregation of extreme opponents (Gul and Pesendorfer, 2010; Kolotilin and Wolitzky, 2024) predict a "gap" in the ideological distribution of elected representatives, with moderate supporters of the redistricting party at one end and extreme opponents at the other. Models without segregation of extreme opponents (Friedman and Holden, 2008) predict more district heterogeneity and a more continuous spectrum of representation. My model, which incorporates endogenous candidate emergence, predicts a significant gap in the distribution of district representatives without relying on segregation of extreme opponents.

Let $\rho^{\omega}_{\mathcal{H}}$ denote the distribution of district representatives given a redistricting plan $\mathcal{H}$ and a shock realization $\omega$. Formally:

**Proposition 6.** *For any optimal plan $\mathcal{H}$ and shock realization $\omega < k$, $\rho^{\omega}_{\mathcal{H}}((-\omega, k)) = 0$.*

Like previous models, optimal plans create many right-leaning districts that elect moderate Republicans. However, opposition voters are never homogeneous in a district. An optimal plan creates districts with the right mix of moderate and extreme Democrats, with two important consequences: first, extreme Democratic candidates win primaries by outnumbering moderates; second, moderate Democrats in these districts are effectively disenfranchised: they lack representatives who share their views and end up voting for the Republican candidate. As it turns out, these disenfranchised moderate Democratic types fall in the interval $(-\omega, k)$, and never end up having their own representation. Hence, my model predicts polarization of representatives as independent of segregation of (extreme) opponents.

## 4.2 Legislative implications for "majority-minority" districts

American federal legislation, including the 1965 Voting Rights Act, mandates that electoral district lines cannot be drawn in such a manner as to improperly dilute minorities' voting power. Since the 1986 Supreme Court decision in Thornburg v. Gingles, such laws have been interpreted as actively requiring the creation of districts where racial and ethnic minorities have the concrete opportunity to elect their own representatives. As a consequence, in the 118th Congress there are 26 congressional districts where Black people constitute a strict majority, and 37 are majority Hispanic or Latino (Klein, 2023). Such districts are called "majority-minority."

While such legislation aims to increase minority representation in Congress, it has sparked debate over its broader political implications. Some argue these districts inadvertently segregate Democratic voters, potentially mirroring aspects of an optimal Republican redistricting strategy. The impact of such legislation on partisan outcomes remains complex and unresolved.

As for the case of Congress polarization, the economic theory literature is divided. One school of thought predicts that segregating extreme opponents is optimal, suggesting majority-minority districts benefit Republicans. The opposing view argues for

more heterogeneous districts, implying that majority-minority districts disrupt Republicans' desired optimum. In my model, imposing a clear homogeneous majority could prevent both Democrats and Republicans from exploiting the endogeneity of electoral incentives.

In some districts, racial or ethnic minorities may constitute a plurality rather than a majority. These areas, known as "minority opportunity" or "non-majority minority" districts, provide these groups the chance to elect their preferred representatives through coalitions with White voters or other minority groups.

Some political analysts argue that Black candidates can win in constituencies where Black voters comprise less than $40-50\%$ of the population. They contend that creating Black-majority districts is unnecessary and may inadvertently limit potential Black political gains (Canon, 2022).

Recent trends, particularly in the South, show people of color winning more seats in majority-white districts (Lublin et al., 2020). However, the success of this strategy largely depends on White Democrats' willingness to support candidates of color. If they are reluctant to do so, a phenomenon known as "White backlash" may occur, potentially benefiting Republican candidates as described in this paper. A similar observation is made by Groll and O'Halloran (2025), who explicitly focus on maximizing minority benefits through redistricting.

# 5 Extensions

## 5.1 Quantile Redistricting

In the previous sections, I relied on party candidates whose positioning rule is a function of party medians. The implication is that the designer can win a district even if there are no supporters inhabiting it. I now consider a scenario where candidates' positions are determined by a quantile, not necessarily the median, of the distribution of preferences within each party. Let $\frac{1}{2} < q < 1$ be the quantile parameter. Under this setting, the Democratic candidate locates at the $q$-quantile of the preference distribution conditional on being affiliated with the Democratic party, while the Republican candidate locates at the $(1-q)$-quantile of the preference distribution conditional on being affiliated with the Republican party. In this setting, as it will become clear soon,

the designer always needs a non-zero measure of supporters in order to win a district. Therefore, it becomes much harder to characterize the solution in general[18]. Thus, for illustrative purposes, I assume $F$ to be uniform on $[\underline{v}, \overline{v}]$.

Call $c^q_{\pi,D}$ and $c^q_{\pi,R}$ the positions taken by $D$ and $R$ in district $\pi$. Then:

$$(c^q_{\pi,D}, c^q_{\pi,R}) = \begin{cases} (v^q_{\pi,D}, k) & if \ \text{supp}(\pi) \subseteq (-\infty, k] \\ (k, v^q_{\pi,R}) & if \ \text{supp}(\pi) \subseteq [k, \infty) \\ (v^q_{\pi,D}, v^q_{\pi,R}) & otherwise \end{cases}$$

where:

$$v^q_{\pi,D} = \inf_a \quad \{a : \ \pi([a, \overline{v}] | v < k) \geqslant q\} \cap \{a : \ \pi([\underline{v}, a] | v < k) \geqslant q\}$$
$$v^q_{\pi,R} = \sup_a \quad \{a : \ \pi([a, \overline{v}] | v \geqslant k) \geqslant 1 - q\} \cap \{a : \ \pi([\underline{v}, a] | v \geqslant k) \geqslant 1 - q\}.$$

Every other detail of the model is the same as in Section 2. The following proposition characterizes the optimal measure of districts won by the designer for each realization of the shock $\omega$, denoted by $V^\omega(q)$.

**Proposition 7.** *There are two cases:*

1. *Suppose $F(k + \omega) \leqslant \frac{1}{2q}$. Then $V^\omega(q) = 1$.*
2. *Suppose $F(k + \omega) > \frac{1}{2q}$. Then $V^\omega(q) = \frac{2q}{2q-1}(1 - F(k + \omega))$.*

In the first case, where the fraction of Democratic voters is less than or equal to $\frac{1}{2q}$, all districts are won by the Republican party. In the second case, where the fraction of Democratic voters is greater than $\frac{1}{2q}$, the designer wins $\frac{2q}{2q-1}(1 - F(k + \omega))$ district, which is strictly more than $2(1-F(k+\omega))$, the districts she would win under exogenous policies. The intuition for the optimal plan is similar to the one in the previous sections. In each district, the redistricter needs to create a wedge between moderate and extreme Democrats, encouraging the emergence of an extreme Democratic candidate and counting on moderate Democrats to vote for a moderate Republican candidate. The following result characterizes optimal redistricting plans:

[18]For starters, the connection to optimal transport requires considering joint distributions with three fixed marginals, rather than just two. Hence, the optimal transport problem itself becomes less tractable.

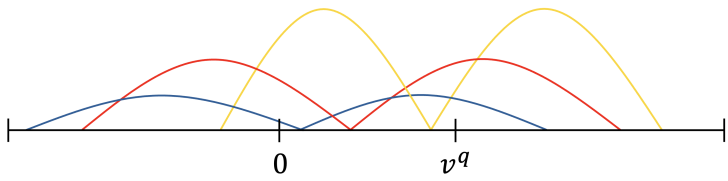



Figure 13: Three-wise positive assortative matching with probability masses of $\frac{1}{2}$, $\frac{1}{2q}$, and $\frac{2q}{2q-1}$.

**Proposition 8.** *Define $v^q = F^{-1}\left(\frac{1}{2q}\right)$. A feasible redistricting plan $\mathcal{H}$ is optimal if and only if, for all $\pi \in supp\ (\mathcal{H})$, there exists $v'_\pi \geqslant v^q$ such that:*

1. $\pi(\{v'_\pi\}) = 1 - \pi(\{v : v < v^q\}) = \frac{2q}{2q-1}$, *and*
2. $|v^q_{\pi,D} - v^m_\pi| \geqslant |v^m_\pi - v'_\pi|$.

This proposition states that an optimal redistricting plan under quantile redistricting must satisfy two conditions for each district $\pi$ in the support of the plan. First, the mass of voters with type exactly equal to $v'_\pi$ should be $\frac{2q}{2q-1}$, and the mass of voters with type below $v_q$ should be $1 - \frac{2q}{2q-1}$. Second, the distance between $v^q_{\pi,D}$, which is the lowest median of the district, and the highest median of the district should be greater than or equal to the distance between the highest median and $v'_\pi$. As it turns out, a type of redistricting plan that I call "three-wise positive assortative" plan complies with the requirements of Proposition 8. Figure 13 illustrates such a plan. Each district contains exactly three voter types, one type below 0, one type between 0 and $v^q$, and another type above $v^q$, with probability masses of $\frac{1}{2}$, $\frac{1}{2q}$, and $\frac{2q}{2q-1}$, respectively. Given shock $\omega$, the designer wins district $\pi$ if and only if $v'_\pi - \omega \geqslant k$. Note that the intuition for the distribution of district representatives works exactly as in Proposition 6. Even under quantile redistricting, the winning candidate is either a Republican with position above $k$, or a Democrat with position below $-\omega$, so that the distribution of district representatives has a gap in $(-\omega, k)$. Comparative statics on the spread of the distribution of voters do not work in the same straightforward way, but are clear nonetheless: the designer benefits from spreading the distribution of voters below 0 and between 0 and $v^q$, and by contracting the distribution of voters above $v^q$. In plain English, the designer wants his opponents to be spread out, while his supporters to be more homogeneous.

## 5.2 Discussion of Alternative Mechasnims

This paper demonstrates that gerrymandering can become more powerful when district composition affects voting behavior. While I have focused on the primary elections channel—where the designer exploits primaries to induce extreme candidates and convert moderate opposition voters—the core mechanism extends to other behavioral responses. For instance, with endogenous turnout, the designer could create districts where moderate Democrats become sufficiently discouraged by extreme Democratic candidates that they abstain rather than vote. The optimal strategy would remain similar: pair extreme and moderate opposition voters to exploit the wedge between them. However, the mechanism would operate through differential participation rather than vote-switching, and the designer would always need a sufficient fraction of Republicans in the district in order to win. Other channels merit exploration as well, including how district composition affects information acquisition, campaign contributions, or grassroots mobilization.

# 6 Conclusions

This paper examines how partisan gerrymandering affects electoral outcomes when candidates' policy positions respond endogenously to district composition. The analysis reveals that optimal gerrymandering strategies differ fundamentally from traditional "pack-and-crack" approaches. While previous literature either predicts extreme opposition voters to be segregated or matched with extreme supporters, my model shows that optimal gerrymandering creates "mismatched slices"—pairing extreme opposition voters with moderate opposition voters. This strategy exploits primary election dynamics to drive opposition candidates toward extreme positions, ultimately allowing the redistricting party to win districts by appealing to moderate voters. These findings suggest that when candidate positions respond to district composition, gerrymandering becomes an even more potent tool than previously understood. By leveraging these electoral dynamics, redistricters can secure victories in more districts than would be possible under traditional approaches that assume fixed voter preferences. This enhanced effectiveness stems from the ability to turn voter diversity into an electoral weakness through strategic district design—a mechanism distinct

from the uncertainty-based explanations in existing work.

These gerrymandering strategies have implications for political representation and polarization. The optimal redistricting plan leads to an ideological gap in Congress, with districts electing either moderate Republicans or extreme Democrats. Notably, this polarization emerges even in heterogeneous districts, challenging the conventional wisdom that political extremism primarily results from voter segregation. The findings also have implications for minority representation, particularly in "minority opportunity" districts where success depends critically on coalition-building with white voters.

Several directions for future research emerge from this analysis, like incorporating idiosyncratic uncertainty about voter preferences or exploring different objective functions for the redistricting party. Other channels leading to endogeneity of voters' behavior are also worth exploring. Across all these possible extensions, the fundamental insight of the paper remains: redistricting has an effect on voting behavior, and designers can exploit it to their advantage.

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# A A Model of Two-Stage District Elections

In this appendix, I develop a model of probabilistic voting at the district level, providing a foundation for the dependence of district candidates' positions on the conditional medians of their respective vote base. While equilibrium policy divergence can be supported under a variety of models, the dependence of such policies on conditional medians is typical of two-stage election mechanisms. I propose one such model in which, like in many others, two forces work in opposing directions to determine equilibrium policies. One such force, a centripetal force, drives positions towards the district median, fueled by the concern voters have with the ability of their first-stage candidate to win general elections. Another force, a centrifugal force, moves positions away from each other, driven by first-stage voters' uncertainty about the exact position of the district median. Similarly to what was suggested, among others, by Owen and Grofman (2006), equilibrium policies tend to be driven towards party medians.

Consider a continuum of voters in a district. Each voter $i$ has policy preferences given by single-peaked utility $u(\cdot, v_i)$, with $v_i \in [\underline{v}, \overline{v}]$ being the voter's unique ideal point. Further, assume that the distribution $\pi$ of $v_i$ admits a unique median, conditional median given $v_i \geqslant k$, and conditional median given $v_i < k$, denoted by $v^m_\pi$, $v^m_{\pi,R}$, and $v^m_{\pi,D}$.

Two-stage elections are held in the district. In the first, primaries stage, there are two Republican and two Democratic candidates. Voters with $v_i \geqslant k$ elect one of the Republican candidates, while voters with $v_i < k$ elect one of the Democratic candidates, by simple majority, with ties broken uniformly at random. In the second stage, general elections, all voters elect a district representative among the two first-stage winners, by simple majority, with ties broken uniformly at random.

Before the first-stage elections, candidates simultaneously announce a policy position and commit to it. They receive a payoff of 1 if they win the second-stage elections and 0 otherwise.

Suppose that voters and candidates do not know the position of the overall district median $v^m_\pi$, and believe it is distributed according to $H \in \Delta([\underline{v}, \overline{v}])$. In addition, candidates know the position of the conditional medians $v^m_{\pi,R}$ and $v^m_{\pi,D}$.

Given voters' single-peaked preferences, the winner of the general elections is the candidate closer to the district median. In the first stage, voters best respond to the

anticipated position of the opposing nominee by voting for the candidate maximizing their expected utility. More formally, given the position of the opposing nominee $y$, a voter with ideal point $t$ votes for the candidate whose position maximizes the following:

$$U(x;y,t) = u(x,t)p(x,y) + u(y,t)(1 - p(x,y)),$$

where $p(x,y)$ is the probability of $x$ winning against $y$ in general elections. As already stated, equilibrium policies in such a model are driven by party medians, but their exact value depends on the functional form of $u$ and $H$. The literare analyses the model various such specifics. I suggest a *linear setting*, where voter preferences are linear and uncertainty is uniform. I prove the following.

**Proposition 9.** *Suppose $u(x,v_i) = -|x - v_i|$ and $H$ is uniform on $[\underline{v},\overline{v}]$. There exists a unique Nash equilibrium where the Republican and Democratic candidates set positions $c_{\pi,R} = v^m_{\pi,R}$ and $c_{\pi,D} = v^m_{\pi,D}$, respectively.*

The advantage of the linear setting is that voters' expected utilities in the primaries turn out to be single-peaked, with maximum reached at each voter's ideal point. Then, it can be argued that any candidate positioning at the party median will win first-stage elections against any other candidate at a different position, irrespective of the opposing party's behavior. Alternatively, Owen and Grofman (2006) consider a model where $u(x,v_i) = e^{-\alpha|x-v_i|}$, for some parameter $\alpha > 0$, and uncertainty is normal around the median, with standard deviation $\sigma$. In their model, voters' expected utility during primary elections is not necessarily single-peaked. Hence, they need to explicitly rule out a particular strategic behavior in which some first-stage voters anticipate that they will prefer the opponents' candidate at general elections and purposefully sabotage their own primaries by voting for an extremist[19]. Under this credible assumption, they prove the following:

**Proposition 10.** (Owen and Grofman, 2006). If $\sigma \geqslant \frac{\max\left\{1-e^{1-\alpha(v^m_\pi - v^m_{\pi,D})}, 1-e^{\alpha(v^m_{\pi,R} - v^m_\pi)}\right\}}{\alpha\sqrt{2\pi}}$,

[19]Under uniform uncertainty, extreme events have a sufficiently high probability of happening, so that voters are deterred from this kind of "political gamble". Indeed, if they sabotage primaries by voting for an extremist, they run the risk of such candidate winning general elections as well.

there exists a unique Nash equilibrium where the Republican and Democratic candidates set positions $c_{\pi,R} = v^m_{\pi,R}$ and $c_{\pi,D} = v^m_{\pi,D}$, respectively.

# B Proofs

## B.1 Proofs of Section 2

**Proof of Proposition 0.** Since the designer wins district $\pi \in \Delta([\underline{v},\overline{v}])$ whenever $v^m_\pi \geqslant k^\star + \omega$, a redistricting plan $\mathcal{H} \in \Delta(\Delta([\underline{v},\overline{v}]))$ can be described by a distribution $\chi \in \Delta([\underline{v},\overline{v}])$ over $v = v^m_\pi$. Using Theorem 2 in Yang and Zentefis (2024), (RPEx) can be stated as:

$$\max_{\chi \in \Delta([\underline{v},\overline{v}])} \int\int \mathbb{1}\left(v - \omega \geqslant k^\star\right) d\chi(v) d\gamma(\omega)$$

$$s.t. \quad \max\{2F(v) - 1, 0\} \leqslant X(v) \leqslant \min\{2F(v), 1\}, \text{ for all } v \in [\underline{v},\overline{v}].$$

Switching the order of integration, it can be rewritten as:

$$\max_{\chi \in \Delta([\underline{v},\overline{v}])} \int G(v - k^\star) d\chi(v)$$

$$s.t. \quad \max\{2F(v) - 1, 0\} \leqslant X(v) \leqslant \min\{2F(v), 1\}, \text{ for all } v \in [\underline{v},\overline{v}].$$

By definition of first order stochastic dominance, and since $G$ is strictly increasing, I have:

$$\int G(v - k^\star) d\chi(v) < \int G(v - k^\star) d\overline{\chi}$$

for any $\chi > \overline{\chi}$, where $\max\{2F(v) - 1, 0\}$ is the CDF associated with $\overline{\chi}$. Hence, the optimal distribution of medians is $X^\star = \max\{2F - 1, 0\}$.

■

**Proof of Proposition 1.** Consider the following definition:

**Definition 1.** *A redistricting plan $\mathcal{H}$ is pairwise if $|\operatorname{supp} \pi| \leqslant 2$ for all $\pi \in \operatorname{supp}(\mathcal{H})$.*

The proof of this proposition relies on the following lemma, which states that for any redistricting plan, there exists a pairwise plan that achieves a weakly higher value, for each realization of the shock $\omega \in \operatorname{supp}(\gamma)$.

**Lemma 1.** *For any feasible plan $\mathcal{H}$, there exists a feasible pairwise plan $\hat{\mathcal{H}}$ such that, for all $\omega \in \text{supp}(\gamma)$:*

$$\int \mathbb{1}\left(c_{\hat{\pi}^\omega} \geqslant k\right) d\hat{\mathcal{H}}(\hat{\pi}) \geqslant \int \mathbb{1}\left(c_{\pi^\omega} \geqslant k\right) d\mathcal{H}(\pi)$$

*Proof.* Take any plan $\mathcal{H} \in \Delta(\Delta([\underline{v}, \overline{v}]))$ such that $\int \pi d\mathcal{H}(\pi) = \phi$. First, for any $\pi \in \text{supp}(\mathcal{H})$, construct a measure $\hat{\mathcal{P}}_\pi \in \Delta(\Delta([\underline{v}, \overline{v}]))$ such that $\int \hat{\pi} d\hat{\mathcal{P}}_\pi(\hat{\pi}) = \pi$, and, for all $\hat{\pi} \in \text{supp}(\hat{\mathcal{P}}_\pi)$, $\text{supp}(\hat{\pi}) = \{v'_{\hat{\pi}}, v''_{\hat{\pi}}\}$, for some $v'_{\hat{\pi}}, v''_{\hat{\pi}} \in \text{supp}(\pi)$ with:

$$v'_{\hat{\pi}} \geqslant v^m_\pi,$$
$$v''_{\hat{\pi}} \leqslant v^m_\pi,$$
$$\hat{\pi}(\{v'_{\hat{\pi}}\}) = \hat{\pi}(\{v''_{\hat{\pi}}\}).$$

Moreover, for any $\hat{\pi}, \hat{\rho} \in \text{supp}(\hat{\mathcal{P}}_\pi)$, let $v'_{\hat{\pi}} > v'_{\hat{\rho}} \implies v''_{\hat{\pi}} \geqslant v''_{\hat{\rho}}$.

Second, construct alternative plan $\hat{\mathcal{H}} \in \Delta(\Delta([\underline{v}, \overline{v}]))$ such that, for any measurable set $A \subseteq \Delta([\underline{v}, \overline{v}])$, $\hat{\mathcal{H}}(A) = \int \hat{\mathcal{P}}_\pi(A) d\mathcal{H}(\pi)$. By construction, $\hat{\mathcal{H}}$ is feasible and pairwise. I now show that:

$$\int \mathbb{1}\left(c_{\hat{\pi}^\omega} \geqslant k\right) d\hat{\mathcal{H}}(\hat{\pi}) \geqslant \int \mathbb{1}\left(c_{\pi^\omega} \geqslant k\right) d\mathcal{H}(\pi).$$

Specifically, I show that, for any $\pi \in \text{supp}(\mathcal{H})$, $\hat{\pi} \in \text{supp}(\hat{\mathcal{P}}_\pi)$, $\omega \in \text{supp}(\gamma)$, I have that $\mathbb{1}\left(c_{\hat{\pi}^\omega} \geqslant k\right) \geqslant \mathbb{1}\left(c_{\pi^\omega} \geqslant k\right)$.

Consider the following cases:

1. If $v^m_{\pi^\omega} \geqslant k$, then $c_{\hat{\pi}^\omega, R} = v^m_{\hat{\pi}^\omega} \geqslant k$, which means that $\mathbb{1}\left(c_{\hat{\pi}^\omega} \geqslant k\right) = 1 \geqslant \mathbb{1}\left(c_{\pi^\omega} \geqslant k\right)$.

2. If $v^m_{\pi^\omega} < k$ and $\text{supp}(\hat{\pi}^\omega) = \{v^m_{\pi^\omega}\}$, it must be that, for all $\hat{\rho} \in \text{supp}(\hat{\mathcal{P}}_\pi)$:

$$v''_{\hat{\rho}} < v''_{\hat{\pi}} = v^m_\pi \implies v'_{\hat{\rho}} \leqslant v'_{\hat{\pi}} = v^m_\pi,$$

   which means that $v''_{\hat{\rho}} < v^m_\pi \implies v'_{\hat{\rho}} = v^m_\pi$, where the equality holds because $v'_{\hat{\rho}} \geqslant v^m_\pi$. Then, it must be that $v^m_{\pi^\omega} = c_{\pi^\omega, D}$ and thus $\mathbb{1}\left(c_{\hat{\pi}^\omega} \geqslant k\right) = \mathbb{1}\left(c_{\pi^\omega} \geqslant k\right) = 0$.

3. If $v^m_{\pi^\omega} < k$, $|\text{supp}(\hat{\pi}^\omega)| = 2$, and $v^m_{\hat{\pi}^\omega} = c_{\hat{\pi}^\omega, R} \geqslant k$, then $\mathbb{1}\left(c_{\hat{\pi}^\omega} \geqslant k\right) = 1 \geqslant \mathbb{1}\left(c_{\pi^\omega} \geqslant k\right)$.

4. If $v^m_{\pi^\omega} < k$, $|\text{supp}(\hat{\pi}^\omega)| = 2$, and $v^m_{\hat{\pi}^\omega} < k$, it must be that $c_{\hat{\pi}^\omega, D} \leqslant c_{\pi^\omega, D}$, $c_{\hat{\pi}^\omega, R} = k \leqslant c_{\pi^\omega, R}$, and $v^m_{\hat{\pi}^\omega} \geqslant v^m_{\pi^\omega}$, so that $\mathbb{1}\left(c_{\hat{\pi}^\omega} \geqslant k\right) \geqslant \mathbb{1}\left(c_{\pi^\omega} \geqslant k\right)$.

□

The proof proceeds in two steps:

1. First, I show that if $\mathcal{H}$ is optimal it must be that, for all $\pi \in \text{supp}(\mathcal{H})$ (except at most for a zero-measure subset), either $|\text{supp}(\pi)| = 1$, or there exist $v' \neq v''$ such that $\pi(\{v'\}) = \pi(\{v''\}) = \frac{1}{2}$.

   Consider plan $\mathcal{H} \in \Delta(\Delta([\underline{v}, \overline{v}]))$ such that $\int d\pi \mathcal{H}(\pi) = \phi$. By Lemma 1, for any $\pi \in \text{supp}(\mathcal{H})$, $\hat{\pi} \in \text{supp}(\hat{\mathcal{P}}_\pi)$, $\omega \in \text{supp}(\gamma)$, I have that $\mathbb{1}(c_{\hat{\pi}^\omega} \geqslant k) \geqslant \mathbb{1}(c_{\pi^\omega} \geqslant k)$. Suppose that, either $|\text{supp}(\pi)| > 2$ or $\text{supp}(\pi) = \{v', v''\}$ with $v' \neq v''$ and $\pi(\{v'\}) \neq \pi(\{v''\})$. I first show that:

$$\int\int \mathbb{1}(c_{\hat{\pi}^\omega} \geqslant k)\, d\hat{\mathcal{P}}_\pi(\hat{\pi}) d\gamma(\omega) > \int \mathbb{1}(c_{\pi^\omega} \geqslant k)\, d\gamma(\omega).$$

   Suppose, without loss of generality[20], that $\text{supp}(\pi) \subseteq (-\infty, k]$. Note that, for any $\hat{\pi}$, $2v^m_{\hat{\pi}^\omega} - c_{\hat{\pi}^\omega, D} - k = 2v^m_{\hat{\pi}} - c_{\hat{\pi}, D} - k - \omega > 0$ for all $\omega \in (v^m_{\hat{\pi}} - k, 2v^m_{\hat{\pi}} - c_{\hat{\pi}, D} - k)$ and $2v^m_{\pi^\omega} - c_{\pi^\omega, D} - k = 2v^m_{\pi} - c_{\pi, D} - k - \omega < 0$ for all $\omega \in (2v_\pi - c_{\pi, D} - k, +\infty)$. Then, for any $\hat{\pi}$ such that $v^m_{\hat{\pi}} - c_{\hat{\pi}, D} > v^m_{\pi} - c_{\pi, D}$, it must be that $2v^m_{\hat{\pi}^\omega} - c_{\hat{\pi}^\omega, D} - k > 0 > 2v^m_{\pi^\omega} - c_{\pi^\omega, D} - k$ for all $\omega \in (2v^m_{\pi} - c_{\pi, D} - k, 2v^m_{\hat{\pi}} - c_{\hat{\pi}, D} - k)$, and thus $\int \mathbb{1}(c_{\hat{\pi}^\omega} \geqslant k)\, d\gamma(\omega) > \int \mathbb{1}(c_{\pi^\omega} \geqslant k)\, d\gamma(\omega)$.

   Excluding the cases $|\text{supp}(\pi)| = 1$ and $|\text{supp}(\pi)| = \{v', v''\}$ with $v' \neq v''$, $\pi(\{v'\}) = \pi(\{v''\})$, it must be that $\hat{\mathcal{P}}_\pi\left(\hat{\pi} : v^m_{\hat{\pi}} - c_{\hat{\pi}, D} > v^m_{\pi} - c_{\pi, D}\right) > 0$. So that:

$$\int\int \mathbb{1}(c_{\hat{\pi}^\omega} \geqslant k)\, d\hat{\mathcal{P}}_\pi(\hat{\pi}) d\gamma(\omega) = \int\int \mathbb{1}(c_{\hat{\pi}^\omega} \geqslant k)\, d\gamma(\omega) d\hat{\mathcal{P}}_\pi(\hat{\pi}) > \int \mathbb{1}(c_{\pi^\omega} \geqslant k)\, d\gamma(\omega),$$

   where the equality follows from $\mathbb{1}(c_{\hat{\pi}^\omega} \geqslant k)$ being bounded.

   Now, suppose that there is a positive-measure subset of $\pi \in \text{supp}(\mathcal{H})$ such that either $|\text{supp}(\pi)| > 2$ or $\text{supp}(\pi) = \{v', v''\}$ with $v' \neq v''$ and $\pi(\{v'\}) \neq \pi(\{v''\})$. I have:

$$\int\int\int \mathbb{1}(c_{\hat{\pi}^\omega} \geqslant k)\, d\hat{\mathcal{P}}_\pi(\hat{\pi}) d\gamma(\omega) d\mathcal{H}(\pi) > \int\int \mathbb{1}(c_{\pi^\omega} \geqslant k)\, d\gamma(\omega) d\mathcal{H}(\pi),$$

   and:

$$\int\int \mathbb{1}(c_{\hat{\pi}^\omega} \geqslant k)\, d\hat{\mathcal{H}}(\hat{\pi}) d\gamma(\omega) = \int\int\int \mathbb{1}(c_{\hat{\pi}^\omega} \geqslant k)\, d\hat{\mathcal{P}}_\pi(\hat{\pi}) d\mathcal{H}(\pi) d\gamma(\omega) >$$
$$> \int\int \mathbb{1}(c_{\pi^\omega} \geqslant k)\, d\gamma(\omega) d\mathcal{H}(\pi).$$

[20] if $\text{supp}(\pi) \nsubseteq (-\infty, 0]$. There exists $\omega^\star$ such that, for all $\omega \geqslant \omega^\star$, $\text{supp}(\pi^\omega) \subseteq (-\infty, 0]$. Then, the reasoning for $\text{supp}(\pi) \subseteq (-\infty, 0]$ applies.

2. I showed that if $\mathcal{H}$ is optimal it must be that, for all $\pi \in \text{supp}(\mathcal{H})$ (except for at most a zero-measure subset), either $|\text{supp}(\pi)| = 1$, or there exist $v', v''$ such that $\pi(\{v'\}) = \pi(\{v''\}) = \frac{1}{2}$. Now I show that it must be that $v' \geqslant v^m$ and $v'' \leqslant v^m$ for all $\pi \in \text{supp}(\mathcal{H})$ (except for at most a zero-measure subset). Suppose there exist $\overline{S} \subseteq \text{supp}(\mathcal{H})$ such that $\mathcal{H}(\overline{S}) > 0$ and for all $\overline{\pi} \in \overline{S}$, $\text{supp}(\overline{\pi}) = \{\overline{v}'_{\overline{\pi}}, \overline{v}''_{\overline{\pi}}\}$ with $\overline{v}'_{\overline{\pi}} \geqslant \overline{v}''_{\overline{\pi}} > v^m$. Since $\int \pi d\mathcal{H}(\pi) = \phi$, there must exist $\underline{S} \subseteq \text{supp}(\mathcal{H})$ such that $\mathcal{H}(\underline{S}) = \mathcal{H}(\overline{S}) > 0$ and for all $\underline{\pi} \in \underline{S}$, $\text{supp}(\underline{\pi}) = \{\underline{v}'_{\underline{\pi}}, \underline{v}''_{\underline{\pi}}\}$ with $\underline{v}''_{\underline{\pi}} \leqslant \underline{v}'_{\underline{\pi}} < v^m$. Construct the measurable set $\hat{S} \subseteq \Delta([\underline{v}, \overline{v}])$ such that, for all $\hat{\pi}' \in \hat{S}$ and $\hat{\pi}'' \in \hat{S}$, there exist $\overline{\pi} \in \overline{S}$ and $\underline{\pi} \in \underline{S}$ such that $\hat{\pi}'(\overline{v}'_{\overline{\pi}}) = \hat{\pi}'(\underline{v}'_{\underline{\pi}}) = \frac{1}{2}$, $\hat{\pi}''(\overline{v}''_{\overline{\pi}}) = \hat{\pi}''(\underline{v}''_{\underline{\pi}}) = \frac{1}{2}$, and vice versa. Note that, by construction, $\mathbb{1}\left(c_{\hat{\pi}'^{\omega}} \geqslant k\right) \geqslant \mathbb{1}\left(c_{\overline{\pi}^{\omega}} \geqslant k\right)$ and $\mathbb{1}\left(c_{\hat{\pi}''^{\omega}} \geqslant k\right) \geqslant \mathbb{1}\left(c_{\underline{\pi}^{\omega}} \geqslant k\right)$ for any $\omega$. Moreover, since $\underline{v}'_{\underline{\pi}} < \overline{v}''_{\overline{\pi}}$, I have that $\mathbb{1}\left(c_{\hat{\pi}'^{\omega}} \geqslant k\right) > \mathbb{1}\left(c_{\overline{\pi}^{\omega}} \geqslant k\right)$ for all $\omega \in (2\overline{v}'_{\overline{\pi}} - \overline{v}''_{\overline{\pi}} - k, 2\overline{v}'_{\overline{\pi}} - \underline{v}'_{\underline{\pi}} - k)$. So:

$$\int \mathbb{1}\left(c_{\hat{\pi}'^{\omega}} \geqslant k\right) d\gamma(\omega) > \int \mathbb{1}\left(c_{\overline{\pi}^{\omega}} \geqslant k\right) d\gamma(\omega).$$

Similarly:

$$\int \mathbb{1}\left(c_{\hat{\pi}''^{\omega}} \geqslant k\right) d\gamma(\omega) > \int \mathbb{1}\left(c_{\underline{\pi}^{\omega}} \geqslant k\right) d\gamma(\omega).$$

Consider the alternative plan $\hat{\mathcal{H}}$, identical to $\mathcal{H}$, but such that $\hat{\mathcal{H}}(\hat{S}) = \mathcal{H}(\overline{S}) + \mathcal{H}(\underline{S})$ and $\hat{\mathcal{H}}(\overline{S}) = \hat{\mathcal{H}}(\underline{S}) = 0$. It then follows that:

$$\int\int \mathbb{1}\left(c_{\pi^{\omega}} \geqslant k\right) d\hat{\mathcal{H}}(\pi) d\gamma(\omega) > \int\int \mathbb{1}\left(c_{\pi^{\omega}} \geqslant k\right) d\mathcal{H}(\pi) d\gamma(\omega).$$

■

**Proof of Theorem 1.** The proof of this result rests on the following lemma.

**Lemma 2.** *There exists a bijection $t : T(\phi', \phi'') \to \Delta_2$.*

*Proof.* Call $\Delta_1 \subseteq \Delta([\underline{v}, \overline{v}])$ the set of district distributions $\pi \in \Delta([\underline{v}, \overline{v}])$ such that there exist $v'_{\pi} \geqslant 0$, $v''_{\pi} \leqslant 0$ with $\pi(\{v'_{\pi}\}) = \pi(\{v''_{\pi}\}) = \frac{1}{2}$. Define the function $s : \Delta_1 \to [\underline{v}, 0] \times [0, \overline{v}]$ such that, for each $\pi \in \Delta_1$, $s(\pi) = (v'_{\pi}, v''_{\pi})$. Note that $s$ is a bijection.

Define the function $t : T(\phi', \phi'') \to \Delta(\Delta_1)$ such that, for all $\tau \in T(\phi', \phi'')$, for all measurable $B \subseteq \Delta_1$, $t(\tau)(B) = \tau(s(B))$. It is easy to see that $t$ is well defined.

First, I show that $t(T(\phi', \phi'')) = \Delta_2$. Indeed, take any $\tau \in T(\phi', \phi'')$. I need to show that $t(\tau)$ is feasible. That is, I show that, for all measurable $A \subseteq [\underline{v}, \overline{v}]$:

$$\int_{\Delta_1} \pi(A) dt(\tau)(\pi) = \phi(A).$$

Note that $\int_{\Delta_1} \pi(A) dt(\tau)(\pi) = \frac{1}{2} \int_{\Delta_1} \pi(A|v \leqslant 0) dt(\tau)(\pi) + \frac{1}{2} \int_{\Delta_1} \pi(A|v \geqslant 0) dt(\tau)(\pi)$. Hence, I show that $\int_{\Delta_1} \pi(A|v \geqslant 0) dt(\tau)(\pi) = \phi'(A)$. The reasoning for $\int_{\Delta_1} \pi(A|v \leqslant 0) dt(\tau)(\pi) = \phi''(A)$ is analogous. By definition of $t$ I have:

$$\int_{\Delta_1} \pi(A|v \leqslant 0) dt(\tau)(\pi) = \int_{\Delta_1} \pi(A|v \leqslant 0) d\tau(s(\pi)).$$

With a change of variable I get:

$$\int_{\Delta_1} \pi(A|v \leqslant 0) d\tau(s(\pi)) = \int_{[0,\overline{v}] \times [\underline{v}, 0]} \mathbb{1}(v' \in A) d\tau(v', v'').$$

Finally:

$$\int_{[0,\overline{v}] \times [\underline{v}, 0]} \mathbb{1}(v' \in A) d\tau(v', v'') = \left( \text{proj}_{[0,\overline{v}]} \# \tau \right)(A) = \phi'(A),$$

where the last equality follows from the definition of transport plan.

Second, I show that $\tau \neq \hat{\tau}$ implies $t(\tau) \neq t(\hat{\tau})$. Since $\tau \neq \hat{\tau}$, there exists measurable $B \subseteq [\underline{v}, 0] \times [0, \overline{v}]$ such that $\tau(B) \neq \hat{\tau}(B)$. Then:

$$t(\tau)(B) = \tau(s(B)) \neq \hat{\tau}(s(B)) = t(\hat{\tau})(B)$$

where the second inequality follows from the fact that $s$ is a bijection. □

I now show that $\tau^\star \in T^\star$ if and only if $\mathcal{H}^\star = t(\tau^\star) \in \Delta_2^\star$.

For any $\mathcal{H} \in \Delta_2$, I have:

$$\int \int \mathbb{1}(c_{\pi^\omega} \geqslant k) d\mathcal{H}(\pi) d\gamma(\omega)$$

$$=$$

$$\int \int \mathbb{1}(v'_\pi - \omega \geqslant k) + \mathbb{1}(v'_\pi - \omega < k) \mathbb{1}\left( v'_\pi - \omega - \frac{v''_\pi - \omega}{2} \geqslant \frac{k}{2} \right) d\mathcal{H}(\pi) d\gamma(\omega).$$

By switching the order of integration and further manipulating, I get:

$$\int\int \mathbb{1}(v'_\pi - \omega \geqslant k) + \mathbb{1}(v'_\pi - \omega < k)\mathbb{1}\left(v'_\pi - \omega - \frac{v''_\pi - \omega}{2} \geqslant \frac{k}{2}\right) d\gamma(\omega) d\mathcal{H}(\pi)$$

$$=$$

$$\int\int \mathbb{1}(v'_\pi - \omega \geqslant k) d\gamma(\omega) + \int \mathbb{1}(v'_\pi - \omega < k)\mathbb{1}\left(v'_\pi - \omega - \frac{v''_\pi - \omega}{2} \geqslant \frac{k}{2}\right) d\gamma(\omega) d\mathcal{H}(\pi)$$

$$=$$

$$\int G(v'_\pi - k) + G\left(2v'_\pi - v''_\pi - k\right) - G(v'_\pi - k) d\mathcal{H}(\pi)$$

$$=$$

$$\int G\left(2v'_\pi - v''_\pi - k\right) d\mathcal{H}(\pi).$$

Using Proposition 1, $\mathcal{H}^\star \in \Delta_2$ is optimal if and only if:

$$\int G\left(2v'_\pi - v''_\pi - k\right) d\mathcal{H}(\pi) \leqslant \int G\left(2v'_\pi - v''_\pi - k\right) d\mathcal{H}^\star(P) \text{ for all} \mathcal{H} \in \Delta_2.$$

With a change of variable:

$$\int G\left(2v'_\pi - v''_\pi - k\right) dt(\tau)(P) = \int G\left(2v'_\pi - v''_\pi - k\right) d\tau(s(P)) = \int G\left(2v'_\pi - v''_\pi - k\right) d\tau(v'_\pi, v''_\pi)$$

$$\leqslant$$

$$\int G\left(2v'_\pi - v''_\pi - k\right) dt(\tau^\star)(\pi) = \int G\left(2v'_\pi - v''_\pi - k\right) d\tau^\star(s(\pi)) = \int G\left(2v'_\pi - v''_\pi - k\right) d\tau^\star(v'_\pi, v''_\pi),$$

for all $\tau \in T(\phi', \phi'')$ and for $\tau^\star = t^{-1}(\mathcal{H}^\star)$. Hence, $\mathcal{H}^\star$ is optimal if and only if $\tau^\star$ is optimal. ■

## B.2 Proofs of Section 3

In this subsection, I borrow results in Santambrogio (2015) and Chiappori et al. (2010) as building blocks to characterize the solution to (OTP). In particular, Lemma 3, Lemma 4, and Lemma 5 are adapted for my context from the above references. Given a function $f : X \to \mathbb{R}$ locally Lipschitz, I define its *superderdifferential* at $x_0 \in X$, $\partial f(x_0)$, to consist of the set of real numbers $\beta$ such that:

$$f(x) \leqslant f(x_0) + \beta(x - x_0) + o(|x - x_0|) \text{ as } x \to x_0,$$

with the error term being allowed to depend on the $x_0$. Note that if the function is differentiable at $x_0$, I have that $\partial f(x_0) = \{f'(x_0)\}$. I provide the following definitions.

**Definition 2.** *$G$ satisfies the twist condition whenever $G$ is locally Lipschitz and $\partial G(a) \cap \partial G(b) = \varnothing$ for all $a \neq b$.*

**Definition 3.** *$G$ satisfies the sub-twist condition whenever $G$ is locally Lipschitz and, for all $a \in \text{supp}(\gamma)$:*

$$|\{b \in \text{supp}(\gamma) : b \neq a,\ \partial G(a) \cap \partial G(b) \neq \varnothing\}| \leqslant 1.$$

**Definition 4.** *For any $\tau \in T(\phi', \phi'')$, $\text{supp}(\tau) \subseteq [0, \overline{v}] \times [\underline{v}, 0]$ is cyclically monotone (CM) if, for every $n \in \mathbb{N}$, every permutation $\sigma$, and every finite family of points $(v_1', v_1''), \ldots, (v_n', v_n'') \in \text{supp}(\tau)$:*

$$\sum_{i=1}^{k} G(2v_i' - v_i'' - k) \geqslant \sum_{i=1}^{k} G(2v_i' - v_{\sigma(i)}'' - k).$$

Then, the following lemmas hold.

**Lemma 3.** *If $G$ satisfies the twist condition, there exists a unique, pure solution to (OTP).*

*Proof.* By Theorem 2 in Chiappori et al. (2010). □

**Lemma 4.** *If $G$ satisfies the sub-twist condition, there exists a unique solution to (OTP).*

*Proof.* By Theorem 3 in Chiappori et al. (2010). □

**Lemma 5.** *If $\tau \in T(\phi', \phi'')$ is a solution to (OTP), then $\text{supp}(\tau)$ is CM.*

*Proof.* By assumption, $G$ is continuous. Hence, the result holds by Theorem 1.38 in Santambrogio (2015). □

I am now ready to prove the results in Section 3.

**Proof of Proposition 2.** Suppose $G$ is strictly convex. Then, for all $x$, either $\partial G(x) = \varnothing$, or $\inf\ \partial G(x') > \sup\ \partial G(x)$ for all $x' > x$. Then, G satisfies the twist condition, and, by Lemma 3, there exists a unique, pure solution $\tau^\star$ to (OTP). Take any $(v', v''), (\tilde{v}', \tilde{v}'') \in \text{supp}(\tau^\star)$, such that $v' > \tilde{v}'$. By Lemma 5, it must be that:

$$G(2v' - v'' - k) + G(2\tilde{v}' - \tilde{v}'' - k) \geqslant G(2v' - \tilde{v}'' - k) + G(2\tilde{v}' - v'' - k).$$

Since $G$ is strictly convex, the function $\Phi(v', v'') = G(2v' - v'' - k)$ is strictly submodular. Suppose that $v'' > \tilde{v}''$. By submodularity:

$$G(2v' - v'' - k) + G(2\tilde{v}' - \tilde{v}'' - k) < G(2v' - \tilde{v}'' - k) + G(2\tilde{v}' - v'' - k).$$

Hence, it must be $v'' \leqslant \tilde{v}''$.

Suppose $G$ is strictly concave. Then, for all $x$, $\sup \partial G(x) < \inf \partial G(x')$ for all $x' > x$. Then, G satisfies the twist condition, and, by Lemma 3, there exists a unique, pure solution $\tau^\star$ to (OTP). Take any $(v', v''), (\tilde{v}', \tilde{v}'') \in \operatorname{supp}(\tau^\star)$, such that $v' > \tilde{v}'$. By Lemma 5, it must be that:

$$G(2v' - v'' - k) + G(2\tilde{v}' - \tilde{v}'' - k) \geqslant G(2v' - \tilde{v}'' - k) + G(2\tilde{v}' - v'' - k).$$

Since $G$ is strictly concave, the function $\Phi(v', v'') = G(2v' - v'' - k)$ is strictly supermodular. Suppose that $v'' < \tilde{v}''$. By supermodularity:

$$G(2v' - v'' - k) + G(2\tilde{v}' - \tilde{v}'' - k) < G(2v' - \tilde{v}'' - k) + G(2\tilde{v}' - v'' - k).$$

Hence, it must be $v'' \geqslant \tilde{v}''$.

Now, suppose $G$ is affine. Then, for any $\tau \in T(\phi', \phi'')$:

$$\int G(2v' - v'' - k) d\tau(v', v'') = G\left(\int (2v' - v'' - k) d\tau(v', v'')\right) =$$
$$= G\left(\int 2v' d\phi' - \int v'' d\phi'' - k\right),$$

by definition of $T(\phi', \phi'')$. Since the objective function is constant over $T(\phi', \phi'')$, I have that $T^\star = T(\phi', \phi'')$.

■

**Proof of Proposition 3.** First, since $G$ is strictly convex below 0 and strictly concave above 0, it satisfies the sub-twist condition, even if it does not necessarily satisfy the twist condition. Hence, by Lemma 4, a solution $\tau$ to (OTP) exists and is unique.

Take any $(v', v''), (\tilde{v}', \tilde{v}'') \in \operatorname{supp}(\tau)$, such that $2v' - v'' - k < 0$ and $2\tilde{v}' - \tilde{v}'' - k < 0$. Suppose that $v' > \tilde{v}'$. By Lemma 5, it must be that:

$$G(2v' - v'' - k) + G(2\tilde{v}' - \tilde{v}'' - k) \geqslant G(2v' - \tilde{v}'' - k) + G(2\tilde{v}' - v'' - k).$$

Suppose that $v'' > \tilde{v}''$. Since $G$ is strictly convex below 0 and such that $G(x) = 1 - G(-x)$, I have:

$$G(2v' - v'' - k) + G(2\tilde{v}' - \tilde{v}'' - k) < G(2v' - \tilde{v}'' - k) + G(2\tilde{v}' - v'' - k).$$

Hence, it must be $v'' \leqslant \tilde{v}''$.

Take any $(v', v''), (\tilde{v}', \tilde{v}'') \in \text{supp}(\tau)$, such that $2v' - v'' - k \geqslant 0$ and $2\tilde{v}' - \tilde{v}'' - k \geqslant 0$. Suppose that $v' > \tilde{v}'$. By Lemma 5, it must be that:

$$G(2v' - v'' - k) + G(2\tilde{v}' - \tilde{v}'' - k) \geqslant G(2v' - \tilde{v}'' - k) + G(2\tilde{v}' - v'' - k).$$

Suppose that $v'' < \tilde{v}''$. Since $G$ is strictly concave above 0, I have:

$$G(2v' - v'' - k) + G(2\tilde{v}' - \tilde{v}'' - k) < G(2v' - \tilde{v}'' - k) + G(2\tilde{v}' - v'' - k).$$

Hence, it must be $v'' \geqslant \tilde{v}''$.

Suppose $\tau(\{v', v'' : \ 2v' - v'' - k \geqslant 0\}) > 0$ and $\tau(\{v', v'' : \ 2v' - v'' - k < 0\}) > 0$. Define the measure $\tau^+$ to be $\tau^+(\cdot) = \tau(\cdot | \{v', v'' : \ 2v' - v'' - k \geqslant 0\})$, and the measure $\tau^-$ to be $\tau^-(\cdot) = \tau(\cdot | \{v', v'' : \ 2v' - v'' - k < 0\})$. Note that $\tau^+$ is positive assortative, while $\tau^-$ is negative assortative. Moreover:

$$\tau(\cdot) = \tau\left(\{v', v'' : \ 2v' - v'' - k \geqslant 0\}\right) \tau^+(\cdot) + \tau\left(\{v', v'' : \ 2v' - v'' - k < 0\}\right) \tau^-(\cdot).$$

Suppose $\tau(\{v', v'' : \ 2v' - v'' - k \geqslant 0\}) = 0$. Then define $\tau^-(\cdot) = \tau(\cdot)$ and note that it is negative assortative.

Suppose $\tau(\{v', v'' : \ 2v' - v'' - k < 0\}) = 0$. Then define $\tau^+(\cdot) = \tau(\cdot)$ and note that it is positive assortative.

Now, because $G$ satisfies the twist condition on $\{v', v'' : \ 2v' - v'' - k < 0\}$ and $\{v', v'' : 2v' - v'' - k \geqslant 0\}$, by Lemma 3, $\tau^+$ and $\tau^-$ are both pure, and thus constitute a positive assortative map and a negative assortative map, respectively. For any $v' \in [0, \overline{v}]$, define the correspondence $C(v') = \{v'' \in [\underline{v}, 0] : (v', v'') \in \text{supp}(\tau)\}$. Note that the restrictions of $C$ on $\{v', v'' : \ 2v' - v'' - k < 0\}$ and $\{v', v'' : \ 2v' - v'' - k \geqslant 0\}$ are functions. Then, path-connectedness of $\text{supp}(\tau^+)$ and $\text{supp}(\tau^-)$ follows directly from the following lemma.

**Lemma 6.** *$C$ is upper hemicontinuous.*

*Proof.* By Theorem 1.38 (strong duality) in Santambrogio (2015):

$$C(v') = \arg\max_{v''} G(2v' - v'' - k) - \psi(v''),$$

where $\psi$ is one of the Kantorovich potentials. By Theorem 1.37 in Santambrogio (2015), and because G is continuous, $\psi$ is also continuous. Then, C is upper hemi-continuous. □

■

**Lemma 7.** *Consider plan* $\tau = \alpha\tau^+ + (1-\alpha)\tau^-$, *in* $T^{\pm}$. *Define* $\Gamma^+ = \text{supp}(\tau^+)$, $\Gamma^- = \text{supp}(\tau^-)$, $\mu^+ = \text{proj}_{[0,\overline{v}]}\#\tau^+$, $\mu^- = \text{proj}_{[0,\overline{v}]}\#\tau^-$, $\nu^+ = \text{proj}_{[\underline{v},0]}\#\tau^+$, $\nu^- = \text{proj}_{[\underline{v},0]}\#\tau^-$. *Consider any set* $\hat{\Gamma} \subseteq \Gamma^-$ *and small* $0 < \epsilon \leqslant 1$. *Construct the measure* $\hat{\tau}^- = \tau^-(\cdot|\hat{\Gamma})$, *with marginals* $\hat{\mu}^- = \text{proj}_{[0,\overline{v}]}\#\hat{\tau}^-$ *and* $\hat{\nu}^- = \text{proj}_{[\underline{v},0]}\#\hat{\tau}^-$. *Define:*

$$\tilde{\tau}^- = \frac{\tau^- - \epsilon\tau^-(\hat{\Gamma})\hat{\tau}^-}{1-\epsilon\tau^-(\hat{\Gamma})},$$

*with marginals* $\tilde{\mu}^- = \frac{\mu^- - \epsilon\tau^-(\hat{\Gamma})\hat{\mu}^-}{1-\epsilon\tau^-(\hat{\Gamma})}$ *and* $\tilde{\nu}^- = \frac{\nu^- - \epsilon\tau^-(\hat{\Gamma})\hat{\nu}^-}{1-\epsilon\tau^-(\hat{\Gamma})}$. *Then, construct* $\tilde{\tau}^+$ *to be positive assortative with marginals* $\tilde{\mu}^+ = \frac{\mu^+ + \frac{1-\alpha}{\alpha}\epsilon\tau^-(\hat{\Gamma})\hat{\mu}^-}{1+\frac{1-\alpha}{\alpha}\epsilon\tau^-(\hat{\Gamma})}$ *and* $\tilde{\nu}^+ = \frac{\nu^+ + \frac{1-\alpha}{\alpha}\epsilon\tau^-(\hat{\Gamma})\hat{\nu}^-}{1+\frac{1-\alpha}{\alpha}\epsilon\tau^-(\hat{\Gamma})}$. *Finally, consider the measure:*

$$\tilde{\tau} = \alpha\left(1 + \frac{1-\alpha}{\alpha}\epsilon\tau^-(\hat{\Gamma})\right)\tilde{\tau}^+ + (1-\alpha)(1-\epsilon\tau^-(\hat{\Gamma}))\tilde{\tau}^-.$$

*If* $\tilde{\tau}$ *is in* $T^{\pm}$, *then:*

$$\int G(2v' - v'' - k)d\tilde{\tau} > \int G(2v' - v'' - k)d\tau.$$

*Proof.* By substitution:

$$\int G(2v' - v'' - k)d\tilde{\tau} =$$
$$= \alpha\left(1 + \frac{1-\alpha}{\alpha}\epsilon\tau^-(\hat{\Gamma})\right)\int G(2v' - v'' - k)d\tilde{\tau}^+ + (1-\alpha)(1-\epsilon\tau^-(\hat{\Gamma}))\int G(2v' - v'' - k)d\tilde{\tau}^-,$$

where:

$$+(1-\alpha)(1-\epsilon\tau^-(\hat{\Gamma}))\int G(2v' - v'' - k)d\tilde{\tau}^- =$$
$$= (1-\alpha)\int G(2v' - v'' - k)d\tau^- - (1-\alpha)\epsilon\tau^-(\hat{\Gamma})\int G(2v' - v'' - k)d\hat{\tau}^-.$$

Hence, the inequality in the statement of the present lemma holds if and only if:

$$\alpha\left(1 + \frac{1-\alpha}{\alpha}\epsilon\tau^-(\hat{\Gamma})\right)\int G(2v' - v'' - k)d\tilde{\tau}^+ - (1-\alpha)\epsilon\tau^-(\hat{\Gamma})\int G(2v' - v'' - k)d\hat{\tau}^- >$$
$$> \alpha\int G(2v' - v'' - k)d\tau^+.$$

Now, define $\hat{\tau}^+ = \tilde{\tau}^+(\cdot|\text{supp}(\hat{\mu}^-) \times [\underline{v}, 0])$ , with marginals $\hat{\mu}^+ = \text{proj}_{[0,\overline{v}]}\#\hat{\tau}^+ = \hat{\mu}^-$ and $\hat{\nu}^+ = \text{proj}_{[\underline{v},0]}\#\hat{\tau}^+$. Morevoer, define $\overline{\tau}^+ = \tilde{\tau}^+(\cdot|\text{supp}(\mu^+) \times [\underline{v}, 0])$ , with marginals $\bar{\mu}^+ = \text{proj}_{[0,\overline{v}]}\#\overline{\tau}^+ = \mu^+$ and $\overline{\nu}^+ = \text{proj}_{[\underline{v},0]}\#\overline{\tau}^+$. Note that:

$$\tilde{\tau}^+ = \frac{\overline{\tau}^+ + \frac{1-\alpha}{\alpha}\epsilon\tau^-(\hat{\Gamma})\hat{\tau}^+}{1 + \frac{1-\alpha}{\alpha}\epsilon\tau^-(\hat{\Gamma})}.$$

By substituting in the inequality:

$$(1-\alpha)\epsilon\tau^-(\hat{\Gamma})\int G(2v' - v'' - k)d\hat{\tau}^+ - (1-\alpha)\epsilon\tau^-(\hat{\Gamma})\int G(2v' - v'' - k)d\hat{\tau}^- >$$
$$> \alpha\int G(2v' - v'' - k)d\tau^+ - \alpha\int G(2v' - v'' - k)d\overline{\tau}^+.$$

To show the inequality holds, note:

1. I have that:

$$\int (2v' - v'' - k)d\hat{\tau}^+ - \int (2v' - v'' - k)d\hat{\tau}^- =$$
$$= -\int v''d\hat{\nu}^+ + \int v''d\hat{\nu}^-,$$

   and:

$$\int (2v' - v'' - k)d\tau^+ - \int (2v' - v'' - k)d\overline{\tau}^+ =$$
$$= -\int v''d\nu^+ + \int v''d\overline{\nu}^+.$$

   Note that it must be that:

$$\nu^+ + \frac{1-\alpha}{\alpha}\epsilon\tau^-(\hat{\Gamma})\hat{\nu}^- = \overline{\nu}^+ + \frac{1-\alpha}{\alpha}\epsilon\tau^-(\hat{\Gamma})\hat{\nu}^+,$$

   so that:

$$\alpha\left(\int (2v' - v'' - k)d\tau^+ - \int (2v' - v'' - k)d\overline{\tau}^+\right) =$$
$$= (1-\alpha)\epsilon\tau^-(\hat{\Gamma})\left(\int (2v' - v'' - k)d\hat{\tau}^+ - \int (2v' - v'' - k)d\hat{\tau}^-\right).$$

2. By assumption $\tilde{\tau}$ is in $T^{\pm}$, so that $2v' - v'' - k \geqslant 0$ for all $(v', v'') \in \text{supp}(\hat{\tau}^+)$ and all $(v', v'') \in \text{supp}(\overline{\tau}^+)$.

3. Putting together 1. and 2. and because $G$ is S-shaped around zero with $G(x) = 1 - G(x)$ delivers the desired inequality.

□

**Proof of Proposition 4.** Suppose $(1-\alpha) \neq 0$ and $2v' - v'' - k > 0$ for all $(v', v'') \in \text{supp}(\tau^+)$, such that $v'' \leqslant v''_{max}$ and $v' \geqslant v'_{max}$. I show that there exist $\hat{\Gamma}$ and $\epsilon$ so that $\tilde{\tau}$ (as defined in Lemma 7) is in $T^{\pm}$. Then, Lemma 7 applies. Consider $v_{max} = (v'_{max}, v''_{max}) = \arg\max_{\text{supp}(\Gamma^-)}(2v' - v'' - k)$. Call $B_\delta(v_{max}) \subseteq \Gamma^-$ a neighborhood of $v_{max}$ in $\Gamma^-$ of radius $\delta$. Take $\epsilon = 1$.

Because $F$ is continuous and strictly increasing on its support, and $2v' - v'' - k > 0$ for all $v', v'' \in \text{supp}(\tau^+)$, there exist $\delta$ such that, for $\hat{\Gamma} = B_\delta(v^\star)$, $2v' - v'' - k \geqslant 0$ for all $v', v'' \in \text{supp}(\hat{\tau}^+)$ and for all $v', v'' \in \text{supp}(\overline{\tau}^+)$, so that $\tilde{\tau}$ is in $T^{\pm}$. The precise argument is as follows.

Denote by $M^+$ and $N^+$ the CDFs associated with the projection of $\tau^+$ on $[0, \overline{v}]$ and $[0, \underline{v}]$, respectively. Call $\delta'$ and $\delta''$ the projections of $\delta$ on $[0, \overline{v}]$ and $[0, \underline{v}]$, respectively. First, note that $v' < v'_{max} - \delta'$ (respectively, $v'' > v''_{max} + \delta''$) is matched to the same $v'' < (F'')^{-1}(M^+)(v'_{max} - \delta')$ (respectively, $v' > (F')^{-1}(N^+)(v''_{max} + \delta'')$) under $\tau^+$ and $\tilde{\tau}^+$. Hence, for the rest of this proof, I focus on the joint events $E'(\delta) = \{v'_{max} - \delta' \leqslant v' \leqslant (F')^{-1}(N^+)(v''_{max} + \delta'')\}$ and $E''(\delta) = \{(F'')^{-1}(M^+)(v'_{max} - \delta') \leqslant v'' \leqslant v''_{max} + \delta''\}$.

For $v' \geqslant v'_{max}$ and $v'' \leqslant v''_{max}$ :

$$M^+(v'|E'(\delta)) =$$
$$= M^+(v'_{max}|E'(\delta)) + (F'(v'|E'(\delta)) - F'(v'_{max}|E'(\delta))\frac{1 - M^+(v'_{max}|E'(\delta))}{1 - F'(v'_{max}|E'(\delta))},$$
$$N^+(v''|E''(\delta)) = F''(v''|E''(\delta))\frac{N^+(v''_{max}|E''(\delta))}{F''(v''_{max}|E''(\delta))}.$$

Both $M^+(\cdot|E'(\delta))$ and $N^+(\cdot|E''(\delta))$ are continuous since $\phi$ is atomless. Assume, without loss of generality, that they are also strictly increasing. If not, is suffices to consider a smaller $\delta$, for which $M^+(v'_{max}|E'(\delta)) = 0$ and/or $1 - N^+(v''_{max}|E''(\delta)) = 0$, and the proof does not change. Then, $M^+(\cdot|E'(\delta))$ and $N^+(\cdot|E''(\delta))$ have a well-defined and continuous inverse, $(M^+)^{-1}$ and $(N^+)^{-1}$ on the interior of their support.

The CDFs associated with the projection of of $\tilde{\tau}^+$ on $[0, \overline{v}]$ and $[0, \underline{v}]$ are simply $F'(\cdot|E'(\delta))$ and $F''(\cdot|E''(\delta))$, respectively. It is easy to check that $M^+(\cdot|E'(\delta))$ dominates $F'(\cdot|E'(\delta))$ in the FOSD sense, while $N^+(\cdot|E''(\delta))$ is dominated by $F''(\cdot|E''(\delta))$ in the FOSD sense.

Under $\tau^+$, each $v' \in E'(\delta)$ is matched to $(N^+)^{-1}(M^+(v'|E'(\delta)))$, and by assumption $2v' - (N^+)^{-1}(M^+(v'|E'(\delta))) - k > 0$. Under $\tilde{\tau}^+$, each $v' \in E'(\delta)$ is matched to

$(F'')^{-1}(F'(v'|E'(\delta)))$. By FOSD:

$$2v' - (N^+)^{-1}(M^+(v'|E'(\delta))) - k > 2v' - (F'')^{-1}(F'(v'|E'(\delta))) - k.$$

However, the difference:

$$[2v' - (N^+)^{-1}(M^+(v'|E'(\delta))) - k] - [2v' - (F'')^{-1}(F'(v'|E'(\delta))) - k]$$

is continuous in $\delta$ and approaches 0 as $\delta'$ and $\delta''$ approach 0. So there exists a small enough $\delta$, such that:

$$2v' - (F'')^{-1}(F'(v'|E'(\delta))) - k > 0.$$

■

**Definition 5** (Median-Preserving Spread). *Distribution $\tilde{\phi} \in \Delta([\underline{v}, \overline{v}])$, with CDF $\tilde{F}$ and median $\tilde{v}^m$, is a median-preserving spread of distribution $\phi \in \Delta([\underline{v}, \overline{v}])$, with CDF $F$ and median $v^m$, if:*

- $\tilde{v}^m = v^m$,
- $F(x) < \tilde{F}(x)$ *for* $x < v^m = \tilde{v}^m$, *and*
- $F(x) > \tilde{F}(x)$ *for* $x > v^m = \tilde{v}^m$.

**Proof of Proposition 5.** Note that $\tilde{\phi}'$ strictly first-order stochastically dominates $\phi'$ and $\phi''$ strictly first-order stochastically dominates $\tilde{\phi}''$. Then:

$$\begin{aligned}
&\max_{\tau \in T(\phi', \phi'')} \int G\left(2v' - v'' - k\right) d\tau(v', v'') \\
&= \int G\left(2v' - v'' - k\right) d\tau^\star(v', v'') \\
&= \int G\left(2v' - v'' - k\right) d\tau^\star(v''|v') d\phi' \\
&< \\
&\int G\left(2v' - v'' - k\right) d\tau^\star(v''|v') d\tilde{\phi}' \\
&\leqslant \\
&\max_{\tau \in T(\tilde{\phi}', \phi'')} \int G\left(2v' - v'' - k\right) d\tau(v', v''),
\end{aligned}$$

where the first inequality follows because $G$ is strictly increasing and by first-order stochastic dominance. Similarly:

$$\max_{\tau \in T(\tilde{\phi}', \phi'')} \int G\left(2v' - v'' - k\right) d\tau(v', v'')$$

$$<$$

$$\max_{\tau \in T(\tilde{\phi}', \tilde{\phi}'')} \int G\left(2v' - v'' - k\right) d\tau(v', v'').$$

So:

$$\max_{\tau \in T(\phi', \phi'')} \int G\left(2v' - v'' - k\right) d\tau(v', v'')$$

$$<$$

$$\max_{\tau \in T(\tilde{\phi}', \tilde{\phi}'')} \int G\left(2v' - v'' - k\right) d\tau(v', v'').$$

## B.3 Proofs of Section 4

**Proof of Proposition 6.** By Proposition 1, any district $\pi$ in an optimal plan $\mathcal{H}$ must be such that $\pi(\{v'\}) = \pi(\{v''\}) = \frac{1}{2}$ for $v' \geqslant 0$ and $v'' \leqslant 0$. Fix shock realization $\omega$. Consider the following cases:

- $v' - \omega \geqslant k$. Then, $c_\pi = v' - \omega \geqslant k$.
- $v' - \omega < k$ and $v' - \omega - \frac{v'' - \omega + k}{2} \geqslant k$. Then $c_\pi = k$.
- $v' - \omega < k$ and $v' - \omega - \frac{v'' - \omega + k}{2} < k$. Then, $c_\pi = v'' - \omega \leqslant -\omega$.

Hence, $Q^\omega_{\mathcal{H}}((-\omega, k)) = 0$.

■

## B.4 Proofs of Section 5.1

**Proof of Proposition 7.** First, note that a necessary, but not sufficient, condition for the designer to win district $\pi$ when the realized shock is $\omega$, is that $\pi^\omega(\{v : v \geqslant k\}) \geqslant \frac{2q-1}{2q}$. Suppose not. Then $\pi^\omega(\{v : v < k\}) > 1 - \frac{2q-1}{2q} = \frac{1}{2q}$. The equilibrium position of the Democratic candidate is $v^q_{\pi^\omega, D}$, the leftmost $q$-quantile of $\pi^\omega$ conditional on $v < k$. Then, it must be that $\pi^\omega(\{v : v \leqslant v^q_{\pi^\omega, D}\}) > \frac{1}{2}$, which means that the district is won by the Democrats. Now, for any redistricting plan $\mathcal{H}$ and any realization of the

schok $\omega$, define by $\chi^\omega \in \Delta([0,1])$ the distribution over $x^\omega = \pi^\omega(\{v : v \geqslant k\})$, that is the distribution of Republican voters across districts. Then, an upper bound on the designer utility is:

$$\int \mathbb{1}\left(x^\omega \geqslant \frac{2q-1}{2q}\right) d\chi^\omega(x^\omega) \leqslant \int \frac{2q}{2q-1} x^\omega d\chi^\omega(x^\omega) = \min\left\{1, \frac{2q}{2q-1}(1-F(k+\omega))\right\},$$

where the equality holds by the law of iterated expectations. To finish the proof, it suffices to show that the above upper bound can always be achieved. Because $F$ is uniform, it is easy to see that the upper bound is achieved by the redistricting plan that matches any $v' \in [v^q, \overline{v}]$ to a $v'' = v^q - \frac{\overline{v}-v'}{2q-1} \in [0, v^q]$ and to a $v''' = -\frac{q}{2q-1}(\overline{v}-v) \in [\underline{v}, 0]$, with respective weights $\frac{2q-1}{2q}$, $\frac{1}{2q}$, and $\frac{1}{2}$.

■

**Proof of Proposition 8.** For any redistricting plan $\mathcal{H}$, call $H(\omega)$ the measure of districts won by the designer when the aggregate shock takes realization $\omega$. By Proposition 7, at any realized shock $\omega$, the designer can win at most measure $\min\left\{1, \frac{2q}{2q-1}(1-F(k+\omega))\right\}$ of districts, so $H(\omega) \leqslant \min\left\{1, \frac{2q}{2q-1}(1-F(k+\omega))\right\}$ at any $\omega$. This implies that any feasible $H$ must satisfy $H(\omega) \leqslant H^\star(\omega)$, where:

$$H^\star(\omega) = \begin{cases} 1 & if \quad \omega \leqslant v^q - k \\ \frac{2q}{2q-1}(1-F(k+\omega)) & if \quad \omega > v^q - k \end{cases}$$

The designer expected utility for any feasible $H$ is then:

$$\int H(\omega) d\gamma(\omega) \leqslant \int H^\star(\omega) d\gamma(\omega),$$

with strict inequality if $H(\omega) \neq H^\star(\omega)$ for any $\omega$. If $H^\star$ is attainable at every $\omega$, $\mathcal{H}$ is optimal if and only if it induces $H^\star$. $H^\star$ is always attainable, as shown by the proof of Proposition 7. This means that $\mathcal{H}$-almost all the districts $\pi$ the designer wins if and only if the shock is at most $\omega$ must satisfy $\pi(\{v : v = \omega\}) = 1 - \pi(\{v : v < v^q\}) = \frac{2q-1}{2q}$. However, since to win a district the designer needs at least $\frac{1}{2}$ of voters to vote for the Republican candidate and $\frac{2q-1}{q} \leqslant \frac{1}{2}$, some voters with $v < v^q$ must vote for $R$. Precisely $\frac{1}{2} - \frac{2q-1}{2q}$ additional voters must prefer $v = \omega$, the Republican candidate, to $v^q_{\pi^\omega,D}$, the Democratic candidate, with $v^q_{\pi^\omega,D}$ being the $q$ - quantile of $\pi^\omega$ conditional on $v < k$. But given that $\pi(\{v : v < k\}) = \frac{1}{2q}$, the Democratic candidate sits at a median of $\pi$. For

him not to win the district it must be that:

$$|v^q_{\pi^\omega,D} - v^m_{\pi^\omega}| \geqslant |v^m_{\pi^\omega} - \omega|.$$

■

## B.5 Proofs of Appendix A

**Proof of Proposition 9.** Consider a Democratic voter with ideal point $t$ (the reasoning is similar for a Republican voter). Given position $y$ of the Republican candidate, the expected utility of electing a first-stage Democratic candidate with position $x$ is:

$$U(x;y,t) = u(x,t)p(x,y) + u(y,t)(1 - p(x,y)),$$

where $p(x,y)$ is the probability of the Democratic candidate winning against the Republican candidate in the general elections. The proof of this result relies on the following lemma.

**Lemma 8.** *$U(x;y,t)$ is strictly increasing for $x < t$, strictly decreasing for $x > t$, and achieves its maximum at $x = t$.*

*Proof.* First, suppose $y > t$. Consider three cases:

**Case $x > y$.** First, note that $U(x;y,t) \leqslant U(y;y,t)$. Moreover, $U(x;y,t)$ is decreasing:

$$U(x;y,t) = -(x-t)\left(1 - \frac{\frac{x+y}{2} - \underline{v}}{\overline{v} - \underline{v}}\right) - (y-t)\left(\frac{\frac{x+y}{2} - \underline{v}}{\overline{v} - \underline{v}}\right),$$

and by taking the first order derivative:

$$U'(x;y,t) = -\left(1 - \frac{\frac{x+y}{2} - \underline{v}}{\overline{v} - \underline{v}}\right) + \frac{x-t}{2(\overline{v} - \underline{v})} - \frac{y-t}{2(\overline{v} - \underline{v})} =$$

$$= -1 + \frac{x+y-2\underline{v}}{2(\overline{v} - \underline{v})} + \frac{x-y}{2(\overline{v} - \underline{v})} = -1 + \frac{x-2\underline{v}}{2(\overline{v} - \underline{v})} < 0$$

**Case $t < x < y$.** $U(x;y,t)$ is decreasing:

$$U(x;y,t) = -(x-t)\left(\frac{\frac{x+y}{2} - \underline{v}}{\overline{v} - \underline{v}}\right) - (y-t)\left(1 - \frac{\frac{x+y}{2} - \underline{v}}{\overline{v} - \underline{v}}\right),$$

and by taking the first order derivative:

$$U'(x;y,t) = -\left(\frac{\frac{x+y}{2} - \underline{v}}{\overline{v} - \underline{v}}\right) - \frac{x-t}{2(\overline{v} - \underline{v})} + \frac{y-t}{2(\overline{v} - \underline{v})} =$$

$$= -\frac{x+y-2\underline{v}}{2(\overline{v} - \underline{v})} + \frac{y-x}{2(\overline{v} - \underline{v})} = \frac{-2x+2\underline{v}}{2(\overline{v} - \underline{v})} < 0$$

**Case** $x < t$**.** $U(x;y,t)$ is increasing:

$$U(x;y,t) = -(t-x)\left(\frac{\frac{x+y}{2} - \underline{v}}{\overline{v} - \underline{v}}\right) - (y-t)\left(1 - \frac{\frac{x+y}{2} - \underline{v}}{\overline{v} - \underline{v}}\right),$$

and by taking the first order derivative:

$$U'(x;y,t) = \left(\frac{\frac{x+y}{2} - \underline{v}}{\overline{v} - \underline{v}}\right) + \frac{x-t}{2(\overline{v} - \underline{v})} + \frac{y-t}{2(\overline{v} - \underline{v})} =$$

$$= \frac{x+y-2\underline{v}}{2(\overline{v} - \underline{v})} + \frac{y+x-2t}{2(\overline{v} - \underline{v})} = \frac{2(x+y-t-\underline{v})}{2(\overline{v} - \underline{v})} > 0$$

Second, suppose $y < t$. Consider three cases:

**Case** $x < y$**.** First, note that $U(x;y,t) < U(y;y,t)$. Moreover, $U(x;y,t)$ is increasing:

$$U(x;y,t) = -(t-x)\left(\frac{\frac{x+y}{2} - \underline{v}}{\overline{v} - \underline{v}}\right) - (t-y)\left(1 - \frac{\frac{x+y}{2} - \underline{v}}{\overline{v} - \underline{v}}\right),$$

and by taking the first order derivative:

$$U'(x;y,t) = \left(\frac{\frac{x+y}{2} - \underline{v}}{\overline{v} - \underline{v}}\right) + \frac{x-t}{2(\overline{v} - \underline{v})} - \frac{y-t}{2(\overline{v} - \underline{v})} =$$

$$= \frac{x+y-2\underline{v}}{2(\overline{v} - \underline{v})} + \frac{x-y}{2(\overline{v} - \underline{v})} = \frac{x-2\underline{v}}{2(\overline{v} - \underline{v})} > 0$$

**Case** $y < x < t$**.** $U(x;y,t)$ is increasing:

$$U(x;y,t) = -(t-x)\left(1 - \frac{\frac{x+y}{2} - \underline{v}}{\overline{v} - \underline{v}}\right) - (t-y)\left(\frac{\frac{x+y}{2} - \underline{v}}{\overline{v} - \underline{v}}\right),$$

and by taking the first order derivative:

$$U'(x;y,t) = \left(1 - \frac{\frac{x+y}{2} - \underline{v}}{\overline{v} - \underline{v}}\right) - \frac{x-t}{2(\overline{v} - \underline{v})} + \frac{y-t}{2(\overline{v} - \underline{v})} =$$

$$= 1 - \frac{x+y-2\underline{v}}{2(\overline{v} - \underline{v})} + \frac{y-x}{2(\overline{v} - \underline{v})} = 1 - \frac{-2x+2\underline{v}}{2(\overline{v} - \underline{v})} > 0$$

**Case** $x > t$**.** $U(x;y,t)$ is decreasing:

$$U(x;y,t) = -(x-t)\left(1 - \frac{\frac{x+y}{2} - \underline{v}}{\overline{v} - \underline{v}}\right) - (t-y)\left(\frac{\frac{x+y}{2} - \underline{v}}{\overline{v} - \underline{v}}\right),$$

and by taking the first order derivative:

$$U'(x;y,t) = -\left(1 - \frac{\frac{x+y}{2} - \underline{v}}{\overline{v} - \underline{v}}\right) + \frac{x-t}{2(\overline{v} - \underline{v})} + \frac{y-t}{2(\overline{v} - \underline{v})} =$$

$$= -1 + \frac{x+y-2\underline{v}}{2(\overline{v} - \underline{v})} + \frac{y+x-2t}{2(\overline{v} - \underline{v})} = -1 + \frac{2(x+y-t-\underline{v})}{2(\overline{v} - \underline{v})} < 0.$$

Finally, note that $U(x;y,t)$ is continuous, so it is reaches its maximum at $x = t$, it is strictly decreasing for $x > t$, and strictly increasing for $x < t$.

□

Given Lemma 8, a Democratic candidate at $v^m_{\pi,D}$ will win first-stage elections against any candidate at a different position (and similarly for Republicans). Indeed, Democratic voters to the right of $v^m_{\pi,D}$, accounting for half of all Democratic voters, prefer a candidate at $v^m_{\pi,D}$ to any other candidate to the left of $v^m_{\pi,D}$. Moreover, Democratic voters to the left of $v^m_{\pi,D}$, prefer a candidate at $v^m_{\pi,D}$ to any other candidate to the right of $v^m_{\pi,D}$. Importantly, this reasoning is independent of the Republican candidate's position $y$. Since positioning at $v^m_{\pi,D}$ ($v^m_{\pi,R}$, respectively) gives a Democratic (Republican) first-stage candidate a positive probability of winning second-stage elections, there exists an equilibrium where both Democratic candidates set at $v^m_{\pi,D}$ and both Republican candidates set at $v^m_{\pi,R}$.

Now, I show there can not be any other equilibrium. First, any situation where first-stage candidates do not tie can not be an equilibrium, because the losing candidate can move to $v^m_{\pi,D}$ ($v^m_{\pi,R}$) and have positive probability of winning second-statge elections. Second, any situation where first-stage candidates do not set the same position can

not be an equilibrium. To see this, note that, in order to have different positions and tie at the same time, one of the Democratic (Republican) first-stage candidate needs to choose a position to the left (right) of $v^m_{\pi,D}$ ($v^m_{\pi,R}$). However, any such position is outside of the support of the median distribution, since candidates know the position of the conditional medians, and is therefore dominated by $v^m_{\pi,D}$ ($v^m_{\pi,R}$). Finally, suppose the two first-stage candidates set at the same position, different from $v^m_{\pi,D}$ ($v^m_{\pi,R}$). Then, there exists small $\epsilon > 0$ such that one of the candidates has a profitable deviation by moving closer to $v^m_{\pi,D}$ ($v^m_{\pi,R}$) by $\epsilon$.

■